

Cancer Statistics, 2011

The Impact of Eliminating Socioeconomic and Racial Disparities on Premature Cancer Deaths

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Abstract

Each year, the American Cancer Society estimates the numbers of new cancer cases and deaths expected in the United States in the current year and compiles the most recent data on cancer incidence, mortality, and survival based on incidence data from the National Cancer Institute, the Centers for Disease Control and Prevention, and the North American Association of Central Cancer Registries and mortality data from the National Center for Health Statistics. A total of 1,596,670 new cancer cases and 571,950 deaths from cancer are projected to occur in the United States in 2011. Overall cancer incidence rates were stable in men in the most recent time period after decreasing by 1.9% per year from 2001 to 2005; in women, incidence rates have been declining by 0.6% annually since 1998. Overall cancer death rates decreased in all racial/ethnic groups in both men and women from 1998 through 2007, with the exception of American Indian/Alaska Native women, in whom rates were stable. African American and Hispanic men showed the largest annual decreases in cancer death rates during this time period (2.6% and 2.5%, respectively). Lung cancer death rates showed a significant decline in women after continuously increasing since the 1930s. The reduction in the overall cancer death rates since 1990 in men and 1991 in women translates to the avoidance of about 898,000 deaths from cancer. However, this progress has not benefitted all segments of the population equally; cancer death rates for individuals with the least education are more than twice those of the most educated. The elimination of educational and racial disparities could potentially have avoided about 37% (60,370) of the premature cancer deaths among individuals aged 25 to 64 years in 2007 alone. Further progress can be accelerated by applying existing cancer control knowledge across all segments of the population with an emphasis on those groups in the lowest socioeconomic bracket. **CA Cancer J Clin** 2011;61:212-236. © 2011 American Cancer Society.

Introduction

Cancer is a major public health problem in the United States and many other parts of the world. Currently, one in 4 deaths in the United States is due to cancer. In this article, we provide the expected numbers of new cancer cases and deaths in 2011, as well as an overview of cancer statistics, including updated incidence, mortality, and survival rates and trends. We also estimate the total number of deaths averted as a result of the decline in cancer death rates since the early 1990s and quantify the impact of eliminating racial and socioeconomic disparities on premature deaths from cancer in 2007.

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TABLE 1. Estimated New Cancer Cases and Deaths by Sex, United States, 2011*

	ESTIMATED NEW CASES			ESTIMATED DEATHS		
	BOTH SEXES	MALE	FEMALE	BOTH SEXES	MALE	FEMALE
All Sites	1,596,670	822,300	774,370	571,950	300,430	271,520
Oral cavity & pharynx	39,400	27,710	11,690	7,900	5,460	2,440
Tongue	12,060	8,560	3,500	2,030	1,320	710
Mouth	11,510	6,950	4,560	1,790	1,130	660
Pharynx	13,580	10,600	2,980	2,430	1,740	690
Other oral cavity	2,250	1,600	650	1,650	1,270	380
Digestive system	277,570	151,540	126,030	139,250	79,020	60,230
Esophagus	16,980	13,450	3,530	14,710	11,910	2,800
Stomach	21,520	13,120	8,400	10,340	6,260	4,080
Small intestine	7,570	3,990	3,580	1,100	610	490
Colon†	101,340	48,940	52,400	49,380	25,250	24,130
Rectum	39,870	22,910	16,960			
Anus, anal canal, & anorectum	5,820	2,140	3,680	770	300	470
Liver & intrahepatic bile duct	26,190	19,260	6,930	19,590	13,260	6,330
Gallbladder & other biliary	9,250	3,990	5,260	3,300	1,230	2,070
Pancreas	44,030	22,050	21,980	37,660	19,360	18,300
Other digestive organs	5,000	1,690	3,310	2,400	840	1,560
Respiratory system	239,320	128,890	110,430	161,250	88,890	72,360
Larynx	12,740	10,160	2,580	3,560	2,840	720
Lung & bronchus	221,130	115,060	106,070	156,940	85,600	71,340
Other respiratory organs	5,450	3,670	1,780	750	450	300
Bones & joints	2,810	1,620	1,190	1,490	850	640
Soft tissue (including heart)	10,980	6,050	4,930	3,920	2,060	1,860
Skin (excluding basal & squamous)	76,330	43,890	32,440	11,980	8,080	3,900
Melanoma-skin	70,230	40,010	30,220	8,790	5,750	3,040
Other nonepithelial skin	6,100	3,880	2,220	3,190	2,330	860
Breast	232,620	2,140	230,480	39,970	450	39,520
Genital system	338,620	250,540	88,080	63,980	34,390	29,590
Uterine cervix	12,710		12,710	4,290		4,290
Uterine corpus	46,470		46,470	8,120		8,120
Ovary	21,990		21,990	15,460		15,460
Vulva	4,340		4,340	940		940
Vagina & other genital, female	2,570		2,570	780		780
Prostate	240,890	240,890		33,720	33,720	
Testis	8,290	8,290		350	350	
Penis & other genital, male	1,360	1,360		320	320	
Urinary system	132,900	90,750	42,150	28,970	19,460	9,510
Urinary bladder	69,250	52,020	17,230	14,990	10,670	4,320
Kidney & renal pelvis	60,920	37,120	23,800	13,120	8,270	4,850
Ureter & other urinary organs	2,730	1,610	1,120	860	520	340
Eye & orbit	2,570	1,270	1,300	240	130	110
Brain & other nervous system	22,340	12,260	10,080	13,110	7,440	5,670
Endocrine system	50,400	12,820	37,580	2,620	1,160	1,460
Thyroid	48,020	11,470	36,550	1,740	760	980
Other endocrine	2,380	1,350	1,030	880	400	480
Lymphoma	75,190	40,880	34,310	20,620	10,510	10,110
Hodgkin lymphoma	8,830	4,820	4,010	1,300	760	540
Non-Hodgkin lymphoma	66,360	36,060	30,300	19,320	9,750	9,570
Myeloma	20,520	11,400	9,120	10,610	5,770	4,840
Leukemia	44,600	25,320	19,280	21,780	12,740	9,040
Acute lymphocytic leukemia	5,730	3,320	2,410	1,420	780	640
Chronic lymphocytic leukemia	14,570	8,520	6,050	4,380	2,660	1,720
Acute myeloid leukemia	12,950	6,830	6,120	9,050	5,440	3,610
Chronic myeloid leukemia	5,150	3,000	2,150	270	100	170
Other leukemia‡	6,200	3,650	2,550	6,660	3,760	2,900
Other & unspecified primary sites†	30,500	15,220	15,280	44,260	24,020	20,240

*Rounded to the nearest 10; estimated new cases exclude basal and squamous cell skin cancers and in situ carcinomas except urinary bladder.

About 57,650 carcinoma in situ of the female breast and 53,360 melanoma in situ will be newly diagnosed in 2011.

†Estimated deaths for colon and rectum cancers are combined.

‡More deaths than cases may reflect lack of specificity in recording underlying cause of death on death certificates or an undercount in the case estimate.

Source: Estimated new cases are based on 1995-2007 incidence rates from 46 states and the District of Columbia, as reported by the North American Association of Central Cancer Registries (NAACCR), representing about 95% of the US population. Estimated deaths are based on US Mortality Data, 1969 to 2007, National Center for Health Statistics, Centers for Disease Control and Prevention.

TABLE 2. Age-standardized Incidence Rates for All Cancers Combined, 2003-2007, and Estimated New Cases* for Selected Cancers by State, United States, 2011

STATE	INCIDENCE RATE†	ALL CASES	FEMALE BREAST	UTERINE CERVIX	COLON & RECTUM	UTERINE CORPUS	LEUKEMIA	LUNG & BRONCHUS	MELANOMA OF THE SKIN	NON-HODGKIN LYMPHOMA	PROSTATE	URINARY BLADDER
Alabama	457.8	25,530	3,700	210	2,310	550	590	4,240	1,260	960	3,680	930
Alaska	463.2	3,090	460	‡	260	80	80	380	90	130	490	130
Arizona	397.5	31,550	4,240	220	2,620	800	780	3,820	1,330	1,220	4,660	1,530
Arkansas	462.7	16,070	2,100	130	1,550	370	420	2,660	500	650	2,400	650
California	440.1	163,480	25,510	1,520	13,880	4,730	4,760	17,660	8,250	7,070	25,030	6,810
Colorado	439.4	22,390	3,390	160	1,780	600	710	2,250	1,130	970	3,920	960
Connecticut	509.4	21,440	3,280	110	1,680	700	520	2,680	1,060	880	3,300	1,050
Delaware	516.0	5,130	810	‡	430	150	120	780	240	200	840	230
Dist. of Columbia§	479.5	2,830	500	‡	240	80	70	360	70	100	580	90
Florida	458.5	113,400	15,330	900	10,180	2,960	3,440	17,150	5,260	4,720	16,780	5,490
Georgia	461.3	44,580	7,030	410	3,940	1,120	1,130	6,410	2,120	1,670	7,360	1,460
Hawaii	431.0	6,710	1,040	50	670	230	170	780	340	230	850	230
Idaho	462.6	7,520	1,030	50	620	210	240	870	340	310	1,320	350
Illinois	488.5	65,610	9,510	570	6,240	2,050	1,870	9,210	2,340	2,640	9,340	2,910
Indiana	470.7	34,050	4,760	260	3,290	1,010	970	5,520	1,410	1,390	4,580	1,440
Iowa	480.7	17,500	2,120	100	1,670	560	580	2,480	890	770	2,590	810
Kansas§	476.0	14,070	1,890	90	1,300	440	430	1,990	710	620	1,870	580
Kentucky	516.2	25,010	3,470	210	2,420	690	650	4,860	1,510	1,040	3,220	1,020
Louisiana	495.1	22,780	2,940	220	2,220	470	620	3,630	630	930	3,640	870
Maine	530.1	8,820	1,280	50	770	300	260	1,400	400	370	1,240	500
Maryland	¶	28,890	4,850	230	2,470	900	700	3,960	1,330	1,130	5,060	1,150
Massachusetts	511.1	37,470	5,640	200	3,000	1,210	970	4,970	1,740	1,550	5,470	1,870
Michigan	500.6	57,010	7,890	360	4,800	1,810	1,630	8,140	2,470	2,330	8,940	2,680
Minnesota	480.5	27,600	3,380	130	2,110	820	820	3,340	880	1,140	4,370	1,100
Mississippi§	471.4	14,990	2,170	150	1,520	320	370	2,430	500	550	2,150	520
Missouri	471.8	32,740	4,100	230	3,150	960	880	5,470	1,310	1,300	4,230	1,370
Montana	459.5	5,690	760	‡	480	150	170	750	190	240	1,020	280
Nebraska	478.1	9,430	1,240	50	930	310	290	1,270	430	430	1,290	410
Nevada	¶	12,800	1,420	110	1,080	290	290	1,510	410	440	1,850	540
New Hampshire	505.8	8,210	1,190	‡	650	260	210	1,110	410	330	1,200	410
New Jersey	509.5	49,080	7,360	430	4,290	1,630	1,360	6,210	2,430	2,140	7,840	2,390
New Mexico	413.1	9,630	1,310	80	820	240	320	980	400	370	1,420	360
New York	491.7	107,260	15,710	960	9,480	3,670	3,070	14,200	3,750	4,650	15,950	5,150
North Carolina	469.5	48,870	7,390	380	4,200	1,280	1,230	7,300	2,300	1,930	7,580	1,900
North Dakota	470.4	3,560	430	‡	340	100	100	420	130	150	600	170
Ohio	470.0	65,060	8,970	480	5,850	2,080	1,690	10,060	2,620	2,660	9,190	2,890
Oklahoma	486.9	18,980	2,680	170	1,800	480	590	3,270	690	850	2,730	760
Oregon	469.8	21,180	3,360	130	1,730	630	560	2,860	1,230	940	3,250	1,020
Pennsylvania	504.3	78,030	10,570	540	7,360	2,620	2,090	10,900	3,240	3,340	11,500	3,920
Rhode Island	517.3	6,090	930	‡	510	200	160	880	270	250	880	320
South Carolina	472.9	25,510	3,710	200	2,100	650	640	3,900	1,200	960	4,230	950
South Dakota	446.6	4,430	590	‡	460	130	140	580	180	190	670	220
Tennessee§	465.8	34,750	5,020	280	3,170	850	930	5,870	1,810	1,410	4,850	1,350
Texas	451.5	105,000	15,070	1,230	9,560	2,670	3,280	13,880	3,970	4,520	15,630	3,670
Utah	404.9	10,530	1,380	70	760	300	320	630	600	440	1,890	400
Vermont	500.0	3,950	590	‡	320	130	100	530	210	160	610	190
Virginia	452.4	38,720	6,480	300	3,420	1,150	940	5,670	1,920	1,520	6,420	1,500
Washington	488.4	35,360	5,630	230	2,720	1,060	1,060	4,540	2,000	1,610	5,470	1,640
West Virginia	497.8	11,080	1,510	80	1,140	360	300	2,080	480	480	1,510	510
Wisconsin	474.7	30,530	4,430	190	2,690	1,060	960	4,020	1,160	1,390	4,900	1,450
Wyoming	444.5	2,680	360	‡	230	70	70	310	110	120	490	130
United States	471.5	1,596,670	230,480	12,710	141,210	46,470	44,600	221,130	70,230	66,360	240,890	69,250

*Rounded to the nearest 10; excludes basal and squamous cell skin cancers and in situ carcinomas except urinary bladder.

†Rates are per 100,000 and age adjusted to the 2000 US standard population.

‡Estimate is fewer than 50 cases.

§This state is not included in the overall US rate because its registry did not achieve high-quality data standards for one or more years during 2003-2007 as determined by the North American Association of Central Cancer Registries (NAACCR).

¶Combined incidence rate is not available.

Source: NAACCR, 2010. Data are collected by cancer registries participating in NCI's SEER Program and CDC's National Program of Cancer Registries. To account for population anomalies caused by Hurricane Katrina in 2005, statistics exclude data for AL, LA, MS, and TX from July 2005 - December 2005.

Note: These model-based estimates are calculated using incidence rates from 46 states and the District of Columbia as reported by NAACCR; they are offered as a rough guide and should be interpreted with caution. State estimates may not add to US total due to rounding and exclusion of states with fewer than 50 cases.

Estimated New Cases*

				Males	Females				
Prostate	240,890	29%			Breast	230,480	30%		
Lung & bronchus	115,060	14%			Lung & bronchus	106,070	14%		
Colon & rectum	71,850	9%			Colon & rectum	69,360	9%		
Urinary bladder	52,020	6%			Uterine corpus	46,470	6%		
Melanoma of the skin	40,010	5%			Thyroid	36,550	5%		
Kidney & renal pelvis	37,120	5%			Non-Hodgkin lymphoma	30,300	4%		
Non-Hodgkin lymphoma	36,060	4%			Melanoma of the skin	30,220	4%		
Oral cavity & pharynx	27,710	3%			Kidney & renal pelvis	23,800	3%		
Leukemia	25,320	3%			Ovary	21,990	3%		
Pancreas	22,050	3%			Pancreas	21,980	3%		
All Sites	822,300	100%				All Sites	774,370	100%	

Estimated Deaths

			Males	Females			
Lung & bronchus	85,600	28%			Lung & bronchus	71,340	26%
Prostate	33,720	11%			Breast	39,520	15%
Colon & rectum	25,250	8%			Colon & rectum	24,130	9%
Pancreas	19,360	6%			Pancreas	18,300	7%
Liver & intrahepatic bile duct	13,260	4%			Ovary	15,460	6%
Leukemia	12,740	4%			Non-Hodgkin lymphoma	9,570	4%
Esophagus	11,910	4%			Leukemia	9,040	3%
Urinary bladder	10,670	4%			Uterine Corpus	8,120	3%
Non-Hodgkin lymphoma	9,750	3%			Liver & intrahepatic bile duct	6,330	2%
Kidney & renal pelvis	8,270	3%			Brain & other nervous system	5,670	2%
All Sites	300,430	100%			All Sites	271,520	100%

FIGURE 1. Ten Leading Cancer Types for the Estimated New Cancer Cases and Deaths By Sex, United States, 2011.

*Estimates are rounded to the nearest 10 and exclude basal and squamous cell skin cancers and in situ carcinoma except urinary bladder.

Material and Methods

Data Sources

Mortality data from 1930 to 2007 in the United States were obtained from the National Center for Health Statistics (NCHS).¹ Incidence data for long-term trends (1975-2007), 5-year relative survival rates, and the lifetime probability of developing cancer were obtained from the Surveillance, Epidemiology, and End Results (SEER) program of the National Cancer Institute (NCI).²⁻⁴ State-specific incidence rates for the years 2003 through 2007 and incidence data (1995-2007) for projecting new cancer cases were obtained from cancer registries that participate in the SEER program or the Centers for Disease Control and Prevention (CDC)'s National Program of Cancer

Registries (NPCR), as reported by the North American Association of Central Cancer Registries (NAACCR).⁵ Population data were obtained from the US Census Bureau.⁶ Cancer cases were classified according to the *International Classification of Diseases for Oncology*.⁷ All incidence and death rates are age-standardized to the 2000 US standard population and expressed per 100,000 population.

Estimated New Cancer Cases and Deaths

The precise number of cancer cases diagnosed each year in the nation and in every state is unknown because cancer registration is incomplete in some states. Furthermore, the most recent year for which incidence and mortality data are available lags 4 years behind the current year due to the time required for

TABLE 3. Age-standardized Death Rates for All Cancers Combined, 2003-2007, and Estimated Deaths* for Selected Cancers by State, United States, 2011

STATE	DEATH RATE†	ALL SITES	BRAIN & OTHER NERVOUS SYSTEM	FEMALE BREAST	COLON & RECTUM	LEUKEMIA	LIVER	LUNG & BRONCHUS	NON-HODGKIN LYMPHOMA	OVARY	PANCREAS	PROSTATE
Alabama	201.5	10,210	210	700	930	350	320	3,210	310	290	600	710
Alaska	180.9	910	‡	70	80	‡	‡	250	‡	‡	60	‡
Arizona	159.6	10,820	290	760	1,020	420	400	2,660	340	330	690	640
Arkansas	202.6	6,460	140	440	580	240	210	2,030	190	150	440	330
California	166.9	56,030	1,480	3,980	4,780	2,200	2,700	12,450	2,050	1,630	4,010	4,330
Colorado	159.8	6,980	210	500	650	300	240	1,690	290	240	480	430
Connecticut	179.3	6,800	150	480	500	260	220	1,750	220	190	550	460
Delaware	198.4	1,930	‡	120	160	60	60	590	50	50	120	110
Dist. of Columbia	198.7	920	‡	80	90	‡	‡	210	‡	‡	70	80
Florida	174.1	40,980	790	2,690	3,370	1,570	1,410	11,460	1,310	1,020	2,610	2,160
Georgia	187.1	15,860	330	1,120	1,420	560	450	4,670	500	440	980	1,080
Hawaii	150.8	2,370	‡	140	220	80	120	580	90	60	180	140
Idaho	170.4	2,570	90	160	210	120	70	630	90	70	200	210
Illinois	191.3	23,140	470	1,830	2,190	900	710	6,420	680	640	1,610	1,310
Indiana	199.4	12,960	340	870	1,090	520	350	4,020	420	350	810	690
Iowa	182.1	6,390	160	380	600	300	170	1,770	290	190	390	410
Kansas	182.7	5,370	140	370	480	300	150	1,600	190	150	340	290
Kentucky	216.5	9,750	190	590	850	320	250	3,420	300	220	550	410
Louisiana	211.5	8,360	210	610	900	300	360	2,480	270	220	540	480
Maine	199.7	3,180	80	170	260	110	90	960	80	80	200	170
Maryland	188.8	10,240	210	800	920	390	380	2,720	300	270	710	770
Massachusetts	186.6	12,910	270	760	980	470	460	3,490	360	370	940	640
Michigan	191.0	20,770	510	1,320	1,670	820	610	5,830	660	560	1,360	1,150
Minnesota	173.7	9,240	230	610	750	390	290	2,470	310	250	610	460
Mississippi	207.7	6,060	150	400	620	220	200	2,010	190	150	360	360
Missouri	197.5	12,700	280	870	1,060	510	390	3,970	450	300	830	540
Montana	179.3	2,000	60	110	170	90	50	570	80	60	120	140
Nebraska	176.5	3,510	90	200	350	140	90	900	140	90	200	280
Nevada	189.1	4,740	120	330	540	100	190	1,290	150	120	320	310
New Hampshire	187.8	2,690	70	190	200	100	80	770	60	60	200	160
New Jersey	186.1	16,370	330	1,260	1,510	610	470	4,160	630	470	1,140	1,100
New Mexico	162.2	3,460	80	240	340	120	160	800	120	90	230	270
New York	171.9	34,350	810	2,450	2,890	1,350	1,310	8,580	1,470	1,000	2,470	1,770
North Carolina	192.3	19,760	340	1,390	1,480	660	520	5,770	550	460	1,200	990
North Dakota	173.3	1,280	‡	80	110	50	‡	310	‡	‡	100	80
Ohio	200.2	24,900	540	1,730	2,170	910	700	7,210	830	600	1,550	1,260
Oklahoma	196.9	7,780	170	530	690	290	230	2,390	280	180	400	350
Oregon	185.8	7,550	210	490	700	280	240	2,110	320	240	540	470
Pennsylvania	193.8	28,560	540	1,970	2,440	1,080	870	7,960	1,090	800	2,070	1,920
Rhode Island	187.5	2,150	50	120	140	90	80	590	50	60	140	80
South Carolina	194.2	9,310	200	660	740	330	280	2,910	300	260	570	550
South Dakota	176.7	1,680	‡	100	150	70	50	450	80	50	110	120
Tennessee	206.0	13,790	340	890	1,170	490	390	4,570	470	330	770	750
Texas	177.5	36,770	830	2,620	3,230	1,410	1,730	9,560	1,060	950	2,260	2,060
Utah	135.7	2,880	100	260	250	140	80	490	100	90	200	230
Vermont	179.7	1,290	‡	100	110	60	‡	360	‡	‡	80	60
Virginia	188.6	14,340	300	1,140	1,270	500	430	4,100	440	410	950	780
Washington	181.2	11,740	380	800	960	490	460	3,090	430	370	790	760
West Virginia	209.8	4,680	100	270	420	140	120	1,480	190	120	220	120
Wisconsin	182.0	11,440	260	690	860	480	340	2,940	390	330	730	600
Wyoming	176.3	1,020	‡	60	110	‡	‡	260	50	‡	70	60
United States	183.8	571,950	13,110	39,520	49,380	21,780	19,590	156,940	19,320	15,460	37,660	33,720

*Rounded to the nearest 10.

†Rates are per 100,000 and age adjusted to the 2000 US standard population.

‡Estimate is fewer than 50 deaths.

Note: State estimates may not add to US total due to rounding and exclusion of states with fewer than 50 deaths.

Source: US Mortality Data, 1969 to 2007, National Center for Health Statistics, Centers for Disease Control and Prevention.

TABLE 4. Probability (%) of Developing Invasive Cancers Within Selected Age Intervals by Sex, United States, 2005-2007*

		BIRTH TO 39	40 TO 59	60 TO 69	70 AND OLDER	BIRTH TO DEATH
All sites†	Male	1.44 (1 in 69)	8.50 (1 in 12)	15.71 (1 in 6)	37.95 (1 in 3)	44.29 (1 in 2)
	Female	2.12 (1 in 47)	9.01 (1 in 11)	10.22 (1 in 10)	26.49 (1 in 4)	37.76 (1 in 3)
Urinary bladder‡	Male	0.02 (1 in 4,693)	0.38 (1 in 262)	0.93 (1 in 107)	3.67 (1 in 27)	3.80 (1 in 26)
	Female	0.01 (1 in 12,116)	0.12 (1 in 836)	0.26 (1 in 390)	0.98 (1 in 102)	1.16 (1 in 87)
Breast	Female	0.48 (1 in 207)	3.75 (1 in 27)	3.45 (1 in 29)	6.53 (1 in 15)	12.15 (1 in 8)
	Male	0.08 (1 in 1,270)	0.91 (1 in 110)	1.46 (1 in 69)	4.38 (1 in 23)	5.30 (1 in 19)
Colorectum	Female	0.08 (1 in 1,272)	0.72 (1 in 138)	1.05 (1 in 95)	4.00 (1 in 25)	4.97 (1 in 20)
	Male	0.17 (1 in 598)	0.22 (1 in 462)	0.33 (1 in 302)	1.20 (1 in 83)	1.52 (1 in 66)
Leukemia	Female	0.13 (1 in 759)	0.15 (1 in 688)	0.20 (1 in 494)	0.78 (1 in 128)	1.10 (1 in 91)
	Male	0.03 (1 in 3,646)	0.93 (1 in 108)	2.29 (1 in 44)	6.70 (1 in 15)	7.67 (1 in 13)
Lung & bronchus	Female	0.03 (1 in 3,185)	0.77 (1 in 130)	1.74 (1 in 57)	4.90 (1 in 20)	6.35 (1 in 16)
	Male	0.15 (1 in 656)	0.64 (1 in 157)	0.74 (1 in 136)	1.85 (1 in 54)	2.73 (1 in 37)
Melanoma of the skin§	Female	0.28 (1 in 353)	0.55 (1 in 181)	0.37 (1 in 267)	0.81 (1 in 123)	1.82 (1 in 55)
	Male	0.13 (1 in 782)	0.44 (1 in 226)	0.60 (1 in 168)	1.73 (1 in 58)	2.30 (1 in 43)
Non-Hodgkin lymphoma	Female	0.08 (1 in 1,179)	0.31 (1 in 318)	0.44 (1 in 229)	1.39 (1 in 72)	1.92 (1 in 52)
	Male	0.01 (1 in 8,517)	2.52 (1 in 40)	6.62 (1 in 15)	12.60 (1 in 8)	16.22 (1 in 6)
Prostate	Female	0.15 (1 in 656)	0.27 (1 in 377)	0.13 (1 in 762)	0.18 (1 in 544)	0.68 (1 in 147)
	Male	0.07 (1 in 1,423)	0.75 (1 in 134)	0.85 (1 in 117)	1.24 (1 in 81)	2.58 (1 in 39)

*For people free of cancer at beginning of age interval.

†All sites excludes basal and squamous cell skin cancers and in situ cancers except urinary bladder.

‡Includes invasive and in situ cancer cases

§Statistics for whites only.

Source: DevCan: Probability of Developing or Dying of Cancer Software, Version 6.5.0. Bethesda, MD: Statistical Research and Applications Branch, National Cancer Institute; 2010.

data collection, compilation, and dissemination. Therefore, we project the expected number of new cancer cases and deaths in the United States in 2011 in order to provide an estimate of the current cancer burden. We projected the estimated number of new malignant cancer cases diagnosed in 2011 using a spatiotemporal model⁸ based on incidence data from 1995 through 2007 from 46 states and the District of Columbia that met the NAACCR's high-quality data standard for incidence, covering about 95% of the US population.⁵ This method accounts for expected delays in case reporting and considers geographic variations in sociodemographic and lifestyle factors, medical settings, and cancer screening behaviors as predictors of incidence. To estimate the numbers of new breast carcinoma in situ and melanoma in situ cases in 2011, we first estimated the number of in situ cases occurring annually from 2000 through 2007 by applying the age-specific incidence rates in the 17 SEER areas to the corresponding US population estimates.^{3,6} We then projected the total number of cases in 2011 based on the annual percent change generated by the joinpoint regression model.⁹

We estimated the number of cancer deaths expected to occur in the United States and in

each state in the year 2011 using the state-space prediction method.¹⁰ Projections are based on underlying cause-of-death from death certificates as reported to the NCHS.¹ This model projects the number of cancer deaths expected to occur in 2011 based on the number that occurred each year from 1969 to 2007 in the United States and in each state separately.

Other Statistics

Incidence rates and trends are adjusted for delays in reporting whenever possible. Delayed adjustment accounts for anticipated future corrections to reported cancer case counts and primarily affects the most recent years of incidence data, especially for cancers such as melanoma, leukemia, and prostate that are frequently diagnosed in outpatient settings; thus, delay-adjusted rates provide the most accurate assessment of trends in the most recent time period. The NCI has developed a method to account for expected reporting delays in SEER registries for all cancer sites combined and many specific cancer sites.¹¹ Long-term incidence and mortality trends for selected cancer sites were previously published in

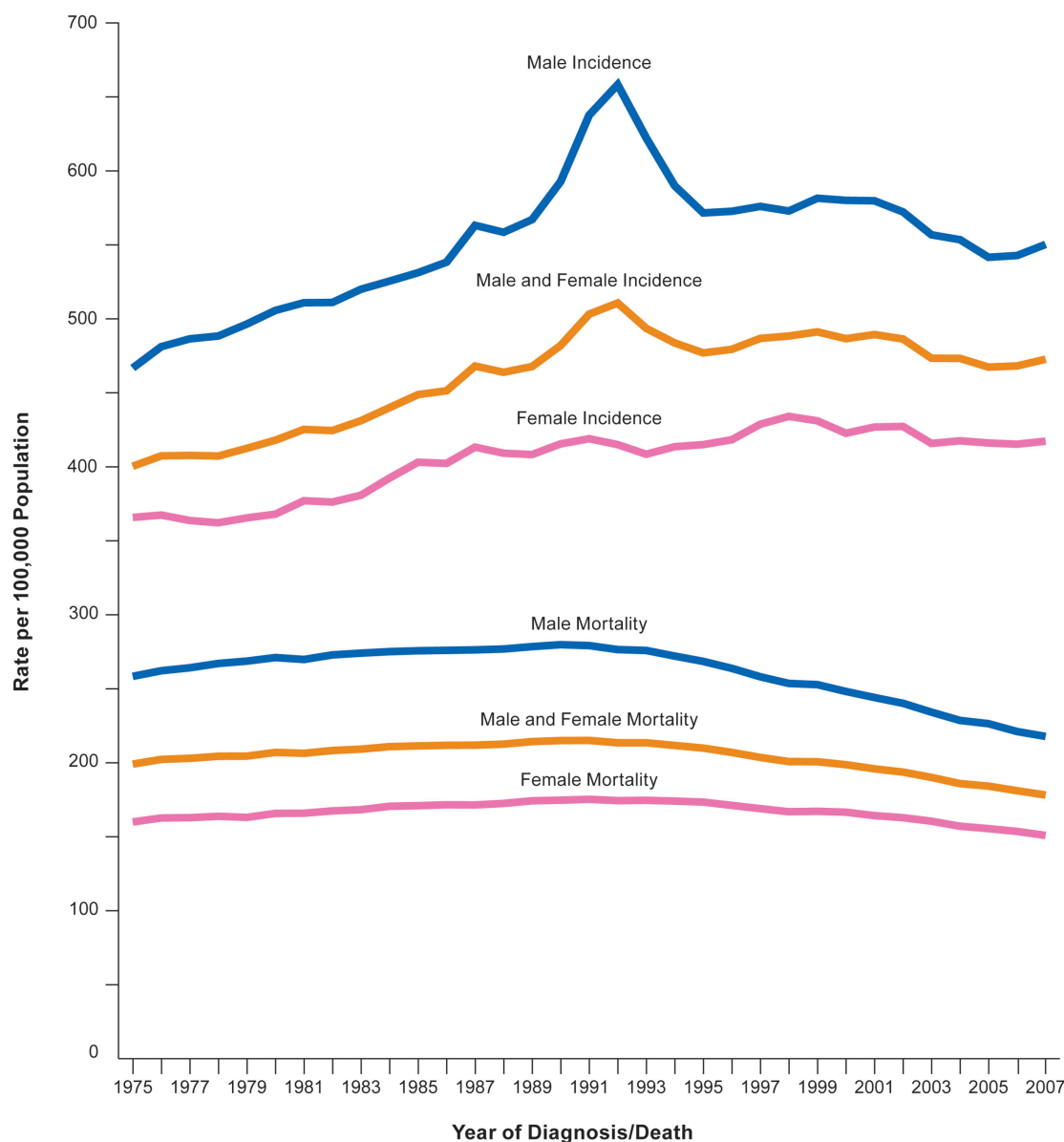


FIGURE 2. Annual Age-Adjusted Cancer Incidence and Death Rates* by Sex, United States, 1975 to 2007.

*Rates are age adjusted to the 2000 US standard population. Incidence rates are adjusted for delays in reporting. Sources: Incidence: Surveillance, Epidemiology, and End Results (SEER) program (available at: www.seer.cancer.gov). Delay-adjusted incidence database: SEER Incidence Delay-Adjusted Rates, 9 Registries, 1975-2007. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Statistical Research and Applications Branch; released April 2010, based on the November 2009 SEER data submission. Mortality: US Mortality Data, 1975 to 2007, National Center for Health Statistics, Centers for Disease Control and Prevention.

the “Annual Report to the Nation on the Status of Cancer.”^{2,12}

The contribution of individual cancer sites to the decrease in cancer death rates was calculated as the proportion of the absolute difference in rates between the peak year (1990 in men and 1991 in women) and 2007 to the total difference in rates for all declining sites combined for each sex. Sites with differences of less than 0.2 per 100,000 were collapsed. The estimated total numbers of cancer deaths avoided in men and women due to the reduction in overall age-standardized cancer death rates through

2007 were calculated by applying the 5-year age-specific cancer death rates in the peak year for the age-standardized cancer death rates (1990 for males and 1991 for females) to the corresponding age-specific populations in the subsequent years through 2007 to obtain the number of expected deaths in each calendar year if death rates had not decreased. We then summed the difference between the number of expected and observed deaths in each age group and calendar year for men and women separately to obtain the total number of cancer deaths avoided over the 17-year interval.

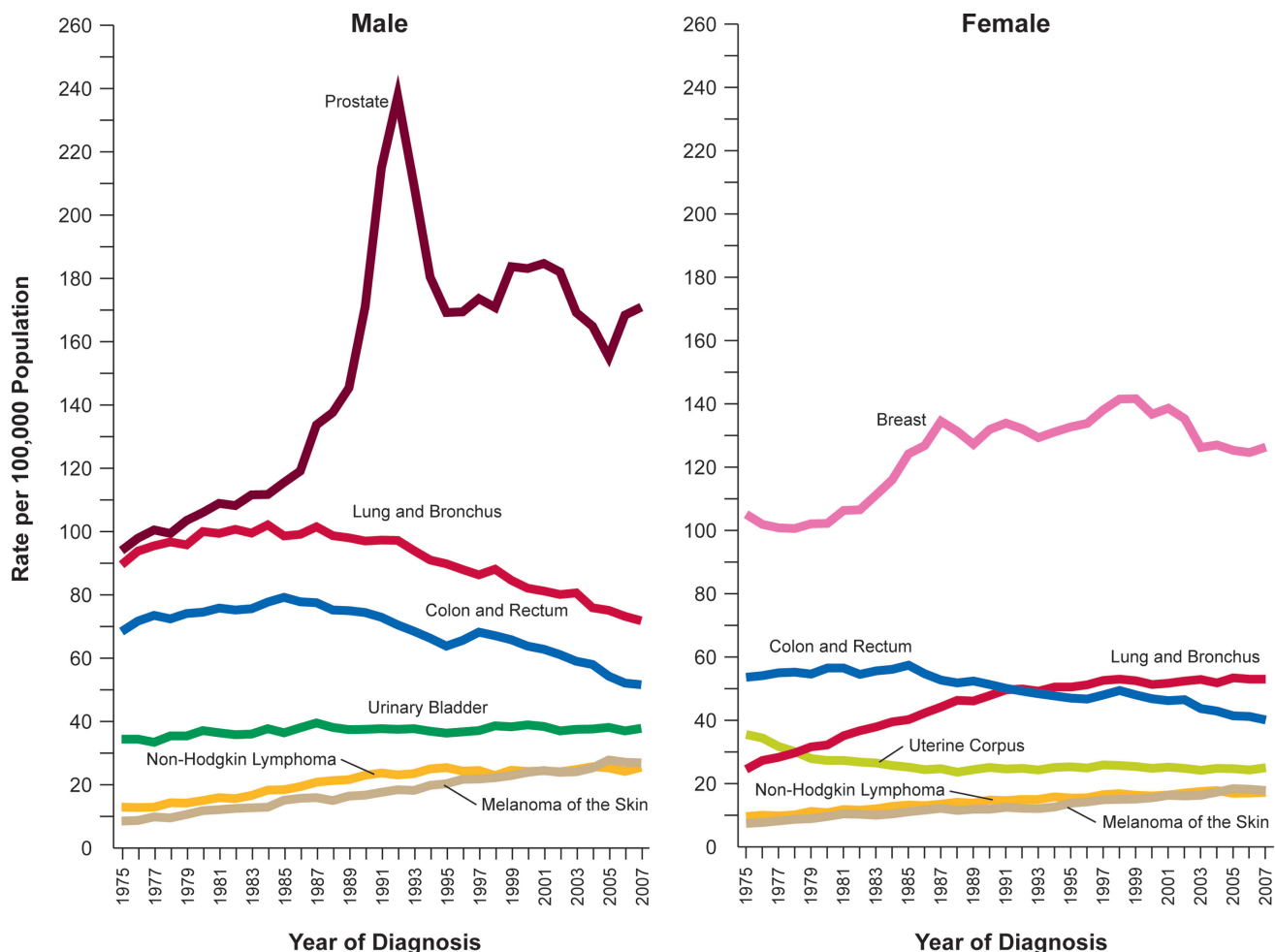


FIGURE 3. Annual Age-Adjusted Cancer Incidence Rates* for Selected Cancers by Sex, United States, 1975 to 2007.

*Rates are age adjusted to the 2000 US standard population and adjusted for delays in reporting. Source: Surveillance, Epidemiology, and End Results (SEER) program (available at: www.seer.cancer.gov). Delay-adjusted incidence database: SEER Incidence Delay-Adjusted Rates, 9 Registries, 1975-2007. Bethesda, MD: National Cancer Institute, Division of Cancer Control and Population Sciences, Surveillance Research Program, Statistical Research and Applications Branch; released April 2010, based on the November 2009 SEER data submission.

Cancer death rates by educational attainment (≤ 12 years of schooling, 13-15 years, and ≥ 16 years) for individuals aged 25 to 64 years in 2007 were calculated using educational attainment information recorded on death certificates and population estimates from the US Census Bureau. Deaths were restricted to those occurring in individuals aged 25 to 64 years because educational attainment is a more reliable index of socioeconomic status in this age group than in older ages¹³; moreover, deaths in younger adults have a larger economic and social impact and are considered premature. Rates were age-standardized to the US 2000 standard population. Rate ratios with corresponding 95% confidence intervals were calculated to compare the least with the most educated groups.¹⁴ The potential number of premature cancer deaths that could have been avoided among individuals aged 25 to 64 years in

2007 by eliminating educational and racial disparities was calculated by applying the age-specific cancer death rates of the most educated non-Hispanic whites in 2007 to all populations. Similarly, we applied the age- and sex-specific death rates of the most educated African Americans to all African Americans and the age-, sex-, and educational attainment-specific death rates of non-Hispanic whites to the corresponding African American population to estimate the numbers of avoidable deaths among the African American population by eliminating educational or racial disparities, respectively.

Selected Findings

Expected Numbers of New Cancer Cases

Table 1 presents estimates of the number of new cases of invasive cancer expected among men and

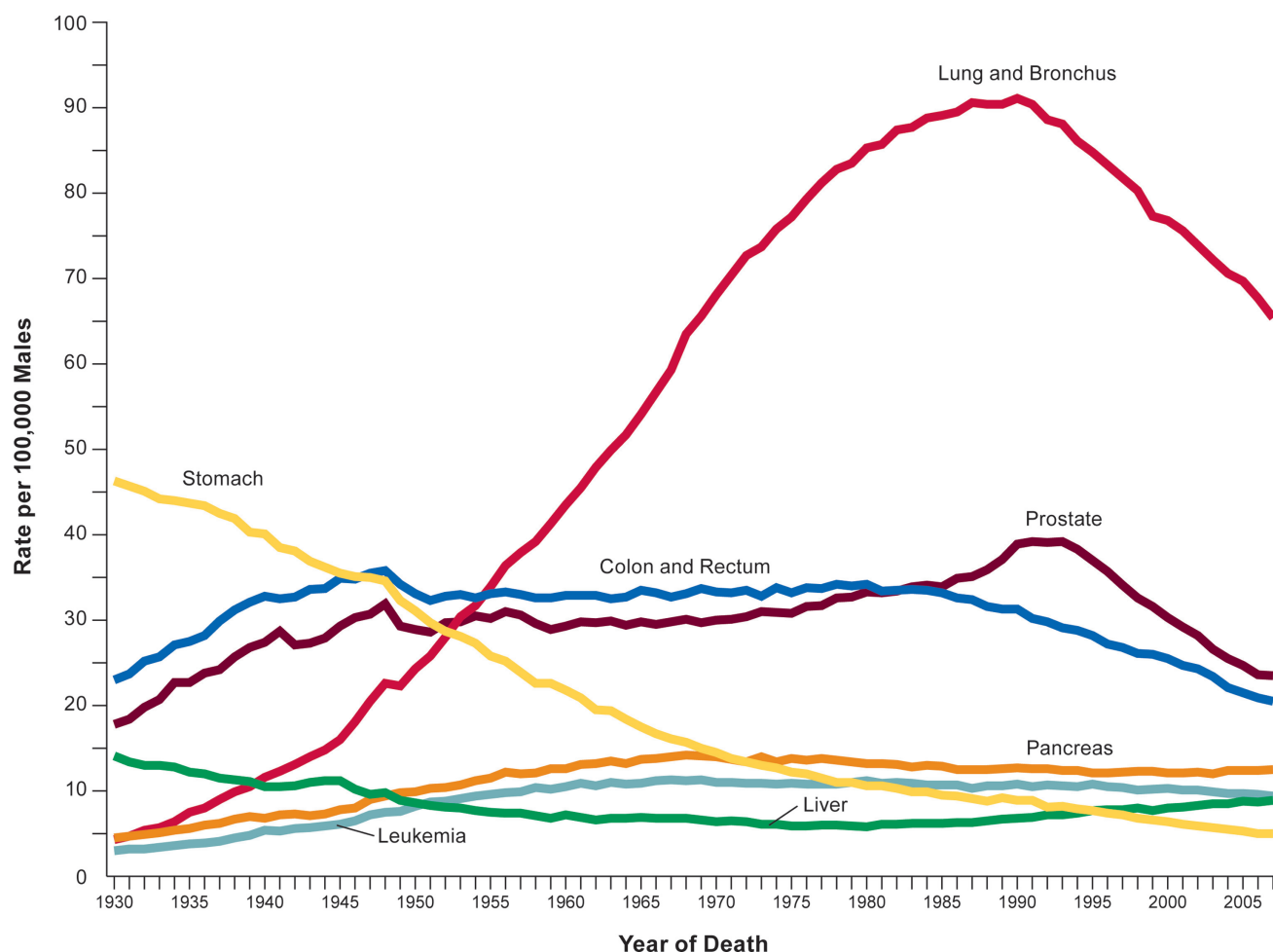


FIGURE 4. Annual Age-Adjusted Cancer Death Rates* Among Males for Selected Cancers, United States, 1930 to 2007.

*Rates are age adjusted to the 2000 US standard population. Due to changes in International Classification of Diseases (ICD) coding, numerator information has changed over time. Rates for cancers of the lung and bronchus, colon and rectum, and liver are affected by these changes. Source: US Mortality Volumes 1930 to 1959, US Mortality Data, 1960 to 2007. National Center for Health Statistics, Centers for Disease Control and Prevention.

women in the United States in 2011. The overall estimate of about 1.6 million new cases does not include carcinoma in situ of any site except the urinary bladder, nor does it include basal cell and squamous cell cancers of the skin. More than 3.5 million unreported cases of basal cell and squamous cell skin cancer (in more than 2 million people), about 57,650 cases of breast carcinoma in situ, and 53,360 cases of melanoma in situ are expected to be newly diagnosed in 2011.¹⁵ The estimated numbers of new cancer cases for each state and selected cancer sites are shown in Table 2.

Figure 1 indicates the most common cancers expected to occur in men and women in 2011. Among men, cancers of the prostate, lung and bronchus, and colorectum will account for about 52% of all newly diagnosed cancers; prostate cancer alone will account for 29% (240,890) of incident cases. The 3 most commonly diagnosed types of cancer

among women in 2011 will be breast, lung and bronchus, and colorectum, accounting for about 53% of estimated cancer cases in women. Breast cancer alone is expected to account for 30% (230,480) of all new cancer cases among women.

Expected Number of Cancer Deaths

Table 1 also shows the expected number of deaths from cancer projected for 2011 for men, women, and both sexes combined. It is estimated that about 571,950 Americans will die from cancer, corresponding to more than 1500 deaths per day. Cancers of the lung and bronchus, prostate, and colorectum in men, and cancers of the lung and bronchus, breast, and colorectum in women continue to be the most common causes of cancer death. These 4 cancers account for almost half of the total cancer deaths among men and women (Fig. 1). Lung cancer surpassed breast cancer as the leading cause of cancer

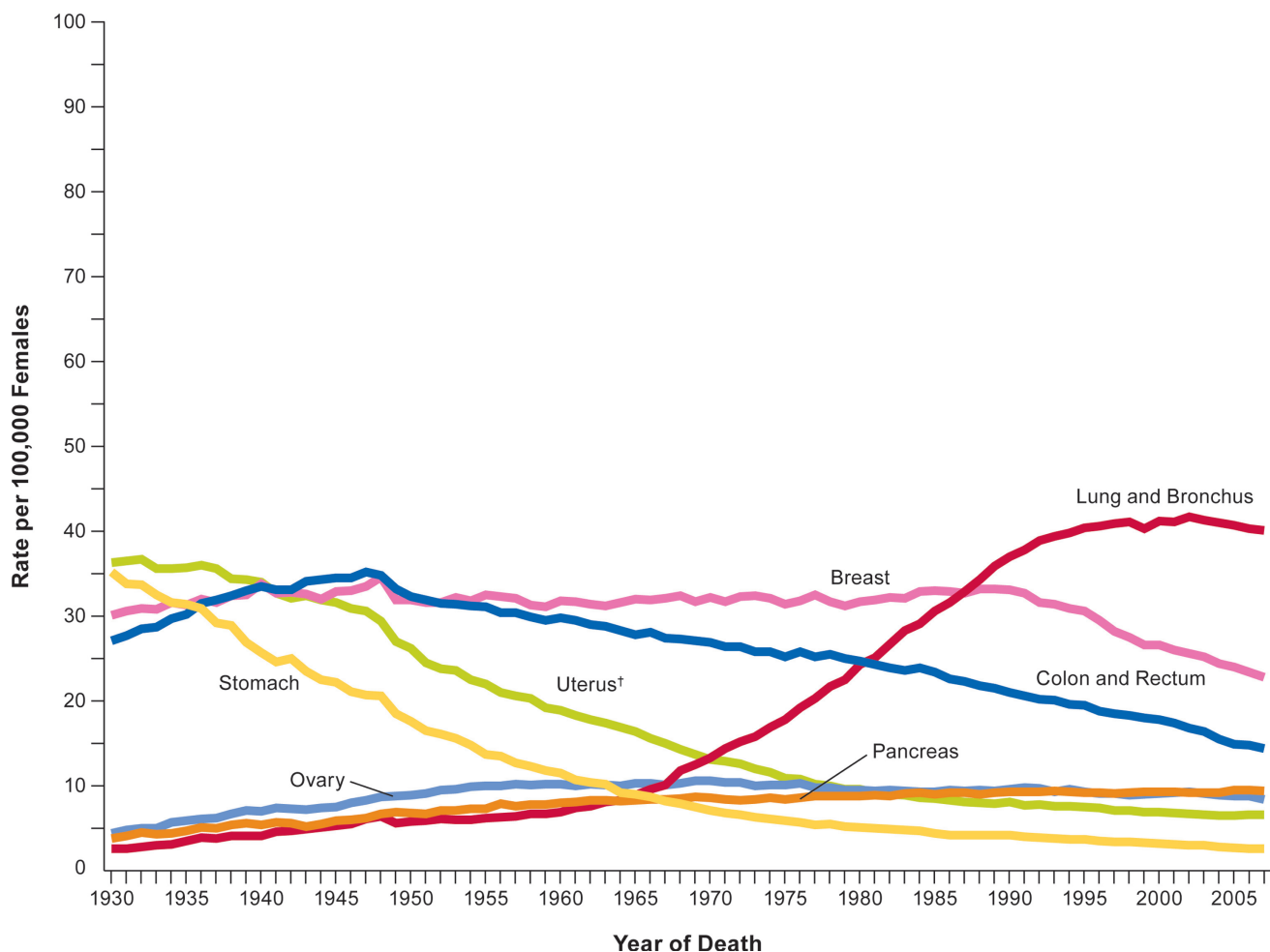


FIGURE 5. Annual Age-Adjusted Cancer Death Rates* Among Females for Selected Cancers, United States, 1930 to 2007.

*Rates are age adjusted to the 2000 US standard population.

†Uterus indicates uterine cervix and uterine corpus. Due to changes in International Classification of Diseases (ICD) coding, numerator information has changed over time. Rates for cancers of the uterus, ovary, lung and bronchus, and colon and rectum are affected by these changes. Source: US Mortality Volumes 1930 to 1959, US Mortality Data, 1960 to 2007. National Center for Health Statistics, Centers for Disease Control and Prevention.

death in women in 1987 and is expected to account for 26% of all female cancer deaths in 2011. Table 3 provides the estimated number of cancer deaths in 2011 by state for selected cancer sites.

Lifetime Probability of Developing Cancer

The lifetime probability of being diagnosed with an invasive cancer is higher for men (44%) than women (38%) (Table 4). However, because of the earlier median age of diagnosis for breast cancer compared with other major cancers, women have a slightly higher probability of developing cancer before age 60 years. These estimates are based on the average experience of the general population and may over- or underestimate individual risk because of differences in exposure (eg, smoking history) and/or genetic susceptibility.

Trends in Cancer Incidence

Figures 2 to 5 depict long-term trends in cancer incidence and death rates for all cancers combined and for selected cancer sites by sex. Table 5 shows long-term incidence (delay-adjusted) and mortality patterns for all cancer sites combined and for the 4 most common cancer sites based on joinpoint regression analysis, which describes trends by fitting annual rates to lines connected at “joinpoints” where trends change in direction or magnitude.^{2,16} Although Table 5 shows a decrease of 1.1% per year from 2000 through 2007 in overall male cancer incidence, the most recent 3 years of delay-adjusted rates indicate a slight increase, which is likely a reflection of an uptick in prostate cancer incidence. Trend analysis based on a larger data set indicates stable incidence rates among men from 2005 to 2007.¹² Overall

TABLE 5. Trends in Cancer Incidence (Delay-Adjusted) and Death Rates for Selected Cancers by Sex, United States, 1975 to 2007

	TREND 1		TREND 2		TREND 3		TREND 4		TREND 5	
	YEARS	APC*	YEARS	APC*	YEARS	APC*	YEARS	APC*	YEARS	APC*
All sites										
Incidence										
Male and female	1975-1989	1.2†	1989-1992	2.8	1992-1995	-2.4	1995-1999	0.9	1999-2007	-0.7†
Male	1975-1989	1.3†	1989-1992	5.2†	1992-1995	-4.8†	1995-2000	0.4	2000-2007	-1.1†
Female	1975-1979	-0.3	1979-1987	1.6†	1987-1995	0.1	1995-1998	1.4	1998-2007	-0.5†
Death										
Male and female	1975-1990	0.5†	1990-1993	-0.3	1993-2001	-1.1†	2001-2007	-1.6†		
Male	1975-1979	1.0†	1979-1990	0.3†	1990-1993	-0.4	1993-2001	-1.5†	2001-2007	-1.9†
Female	1975-1990	0.6†	1990-1994	-0.1	1994-2002	-0.8†	2002-2007	-1.5†		
Lung & bronchus										
Incidence										
Male	1975-1982	1.4†	1982-1991	-0.4	1991-2007	-1.8†				
Female	1975-1982	5.5†	1982-1991	3.5†	1991-2007	0.4†,‡				
Death										
Male	1975-1978	2.4†	1978-1984	1.2†	1984-1991	0.3†	1991-2005	-1.9†	2005-2007	-3.0†
Female	1975-1982	6.0†	1982-1990	4.2†	1990-1995	1.7†	1995-2003	0.3†	2003-2007	-0.9†
Colorectum										
Incidence										
Male	1975-1985	1.1†	1985-1991	-1.2†	1991-1995	-3.2†	1995-1998	2.1	1998-2007	-3.1†
Female	1975-1985	0.3	1985-1995	-1.9†	1995-1998	1.9	1998-2007	-2.2†		
Death										
Male	1975-1984	-0.1	1984-1990	-1.4†	1990-2002	-2.0†	2002-2005	-4.3†	2005-2007	-2.1
Female	1975-1984	-1.0†	1984-2001	-1.8†	2001-2007	-3.2†				
Female breast										
Incidence	1975-1980	-0.5	1980-1987	3.9†	1987-1995	-0.1	1995-1998	2.7	1998-2007	-1.6†
Death	1975-1990	0.4†	1990-2007	-2.2†						
Prostate										
Incidence	1975-1988	2.6†	1988-1992	16.5†	1992-1995	-11.5†	1995-2000	2.1	2000-2007	-1.7†
Death	1975-1987	0.9†	1987-1991	3.0†	1991-1994	-0.5	1994-2005	-4.1†	2005-2007	-2.6†

*Annual percent change (APC) based on incidence (SEER 9 areas) and mortality rates age adjusted to the 2000 US standard population.

†The APC is significantly different from zero.

‡Analysis of the SEER 13 areas published in Kohler et al.¹² shows a significant decrease of 0.3% per year from 2003-2007.

Note: Trends were analyzed by Joinpoint Regression Program, version 3.4.3, with a maximum of four joinpoints (ie, five line segments).

Source: Altekruse, et al.²

cancer incidence rates in females decreased by about 0.5% per year from 1998 through 2007. Incidence trends are decreasing for the 4 major cancer sites except for lung cancer in women, in whom rates are still increasing at a slow rate from 1991 through 2007. However, analysis based on a larger geographic area showed that lung cancer incidence rates are decreasing in the most recent 5 years,¹² which is consistent with the most recent mortality trend based on national-level data. The lag in the temporal trend of lung cancer rates in women compared with men reflects historical differences in cigarette smoking between men and women; cigarette smoking in women peaked about 20 years later than in men. Recent rapid declines in colorectal cancer incidence rates largely reflect increases in screening that can detect and remove precancerous polyps.^{17,18} Although long-term incidence data indicate a decrease in prostate cancer since 2000, delay-adjusted

rates in the most recent time period have increased from 155 (per 100,000) in 2005 to 171 in 2007.² The decrease in the breast cancer incidence rate since 1998 likely reflects the large decline in the use of menopausal hormone therapy among postmenopausal women beginning in 2001, and it may also reflect delayed diagnosis due to decreased mammography utilization from 2003 to 2005.^{19,20} However, close inspection of incidence data by individual year shows that after dramatically decreasing from 2002 to 2003, incidence rates from 2003 to 2007 remained relatively unchanged.²¹ This may support the hypothesis that postmenopausal hormones may be acting as promoters rather than initiators of breast cancer.²⁰

Trends in Cancer Mortality

Cancer replaced heart disease as the leading cause of death among men and women aged younger than 85

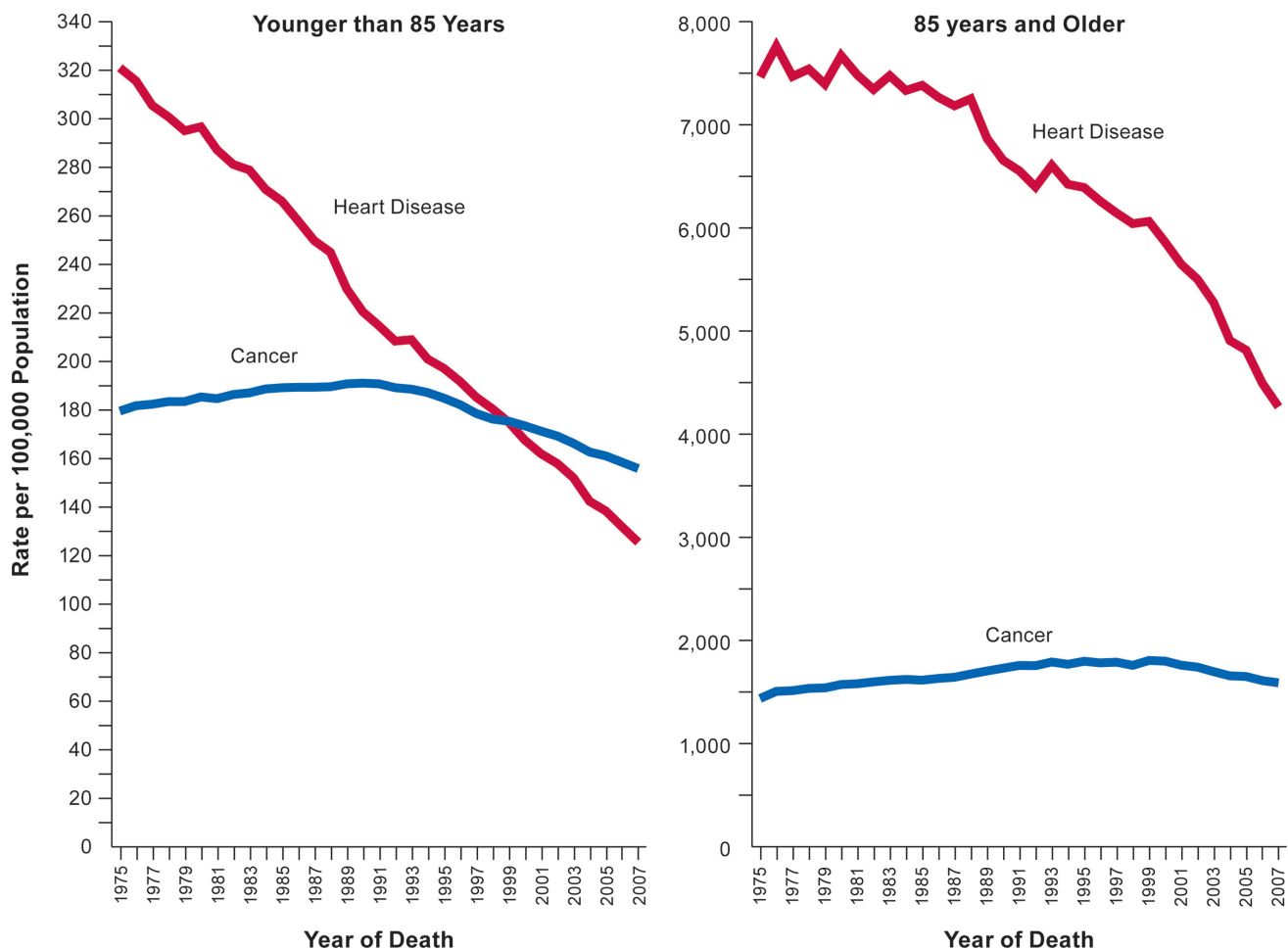


FIGURE 6. Death Rates* for Cancer and Heart Disease for Ages Younger Than 85 Years and 85 Years and Older, 1975 to 2007.
 *Rates are age adjusted to the 2000 US standard population. Source: US Mortality Data, 1975 to 2007. National Center for Health Statistics, Centers for Disease Control and Prevention.

years in 1999 (Fig. 6). The overall cancer death rate decreased by 1.9% per year from 2001 through 2007 in males and by 1.5% in females from 2002 through 2007, compared with smaller declines of 1.5% per year in males from 1993 through 2001 and 0.8% per year in females from 1994 through 2002 (Table 5). Notably, the lung cancer mortality rate in women has begun to decline for the first time in recorded history and more than a decade later than the decline began in men. Mortality rates have continued to decrease for colorectal, female breast, and prostate cancers. Table 6 shows the contribution of individual cancer sites to the total decrease in overall cancer death rates by comparing rates in the current year with those in the peak year—1990 for men and 1991 for women—for all sites combined. Between 1990/1991 and 2007, cancer death rates decreased by 22.2% in men and by 13.9% in women. Among men, the

reduction in death rates for lung, prostate, and colorectal cancers accounted for nearly 80% of the total decrease in the cancer death rate, while among women, the reduction in death rates for breast and colorectal cancers accounted for almost 60% of the decrease. Lung cancer in men and breast cancer in women each account for more than one-third of the sex-specific decreases in cancer death rates. The decrease in lung cancer death rates among men since 1990 is due to the reduction in tobacco use over the past 50 years, while the decrease in death rates for female breast, colorectal, and prostate cancer largely reflects improvements in early detection and/or treatment.

Figure 7 shows the total number of cancer deaths avoided since death rates began to decrease in 1991 in men and in 1992 in women. About 898,000 cancer deaths (649,300 in men and 248,600 in women) were averted during the time interval from 1991/1992 through 2007.

TABLE 6. The Contribution of Individual Cancer Sites to the Decrease in Cancer Death Rates, 1990 to 2007

MALE	DEATH RATE (PER 100,000)		CHANGE		% CONTRIBUTION†
	1990*	2007	ABSOLUTE	%	
All malignant cancers	279.82	217.79	-62.03	-22.17	
Decreasing					
Lung & bronchus	90.56	65.23	-25.33	-27.97	38.5
Prostate	38.56	23.50	-15.06	-39.06	22.9
Colorectum	30.77	20.05	-10.72	-34.84	16.3
Stomach	8.86	5.01	-3.85	-43.45	5.9
Oral cavity & pharynx	5.61	3.85	-1.76	-31.37	2.7
Non-Hodgkin lymphoma	9.97	8.29	-1.68	-16.85	2.6
Leukemia	10.71	9.44	-1.27	-11.86	1.9
Larynx	2.97	2.05	-0.92	-30.98	1.4
Brain & other nervous system	5.97	5.10	-0.87	-14.57	1.3
Myeloma	4.83	4.39	-0.44	-9.11	0.7
Urinary bladder	7.97	7.56	-0.41	-5.14	0.6
Kidney & renal pelvis	6.16	5.79	-0.37	-6.01	0.6
Hodgkin lymphoma	0.85	0.50	-0.35	-41.18	0.5
Other decreasing	38.66	35.89	-2.77	-7.17	4.2
Total	262.45	196.65	-65.80		100.0
Increasing					
Liver & intrahepatic bile duct	5.27	7.92	2.65	50.28	
Esophagus	7.16	7.67	0.51	7.12	
Melanoma of the skin	3.80	3.98	0.18	4.74	
Other increasing	0.84	1.29	0.45	53.57	
Total	17.07	20.86	3.79		
No change					
Bones & joints	0.55	0.55	0.00	0.00	
FEMALE	DEATH RATE (PER 100,000)		CHANGE		% CONTRIBUTION†
	1991*	2007	ABSOLUTE	%	
All malignant cancers	175.30	150.87	-24.43	-13.93	
Decreasing					
Breast	32.69	22.84	-9.85	-30.13	35.2
Colorectum	20.3	14.15	-6.15	-30.30	22.0
Non-Hodgkin lymphoma	6.74	5.19	-1.55	-23.00	5.5
Stomach	4.01	2.57	-1.44	-35.91	5.1
Ovary	9.51	8.22	-1.29	-13.56	4.6
Cervix uteri	3.49	2.42	-1.07	-30.66	3.8
Leukemia	6.32	5.27	-1.05	-16.61	3.8
Brain & other nervous system	4.11	3.47	-0.64	-15.57	2.3
Oral cavity & pharynx	2.03	1.42	-0.61	-30.05	2.2
Myeloma	3.26	2.81	-0.45	-13.80	1.6
Gallbladder	1.09	0.73	-0.36	-33.03	1.3
Kidney & renal pelvis	2.95	2.63	-0.32	-10.85	1.1
Esophagus	1.81	1.58	-0.23	-12.71	0.8
Larynx	0.63	0.42	-0.21	-33.33	0.8
Other decreasing	21.00	18.25	-2.75	-13.10	9.8
Total	119.94	91.97	-27.97	-23.32	100.0
Increasing					
Lung & bronchus‡	37.61	39.98	2.37	6.31	
Liver & intrahepatic bile duct	2.52	3.24	0.73	28.83	
Pancreas	9.28	9.37	0.09	0.97	
Total	49.40	52.59	3.19		
No change					
Uterine corpus	4.18	4.18	0.00	0.00	

*Death rates for cancer peaked in 1990 in men and in 1991 in women.

†This calculation is based on each cancer site's contribution to the increasing or decreasing portion of the total cancer death rate, depending on the individual site's trend; it does not represent the contribution to the net decrease in cancer death rates.

‡Lung cancer death rates in women are decreasing in the most recent time period.

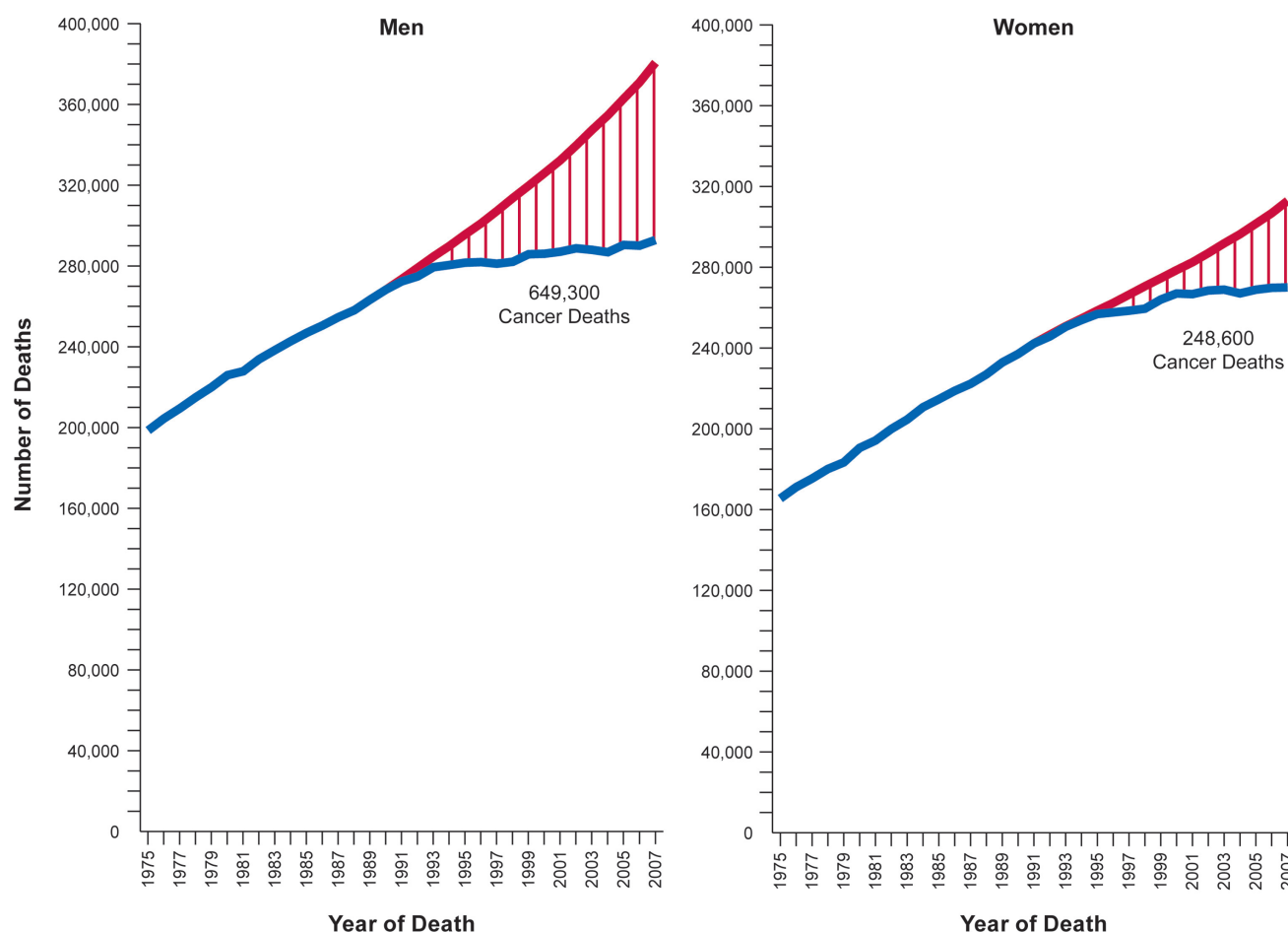


FIGURE 7. Total Number of Cancer Deaths Avoided From 1991 to 2007 in Men and From 1992 to 2007 in Women.

The blue line represents the actual number of cancer deaths recorded in each year, and the bold red line represents the expected number of cancer deaths if cancer mortality rates had remained the same since 1990 and 1991.

Regional Variations in Cancer Rates

Tables 7 and 8 depict cancer incidence and death rates for selected cancer sites by state. Lung cancer shows by far the largest geographic variation in cancer occurrence; lung cancer incidence and death rates in men and women are 3-fold higher in Kentucky (the state with the highest rates) than in Utah (the state with the lowest rates). This variation reflects the large historical and continuing differences in smoking prevalence among states. Utah ranks lowest in adult smoking prevalence and Kentucky among the highest. In contrast, state variations in rates for other cancer sites are smaller in both absolute and proportionate terms. For cancers that can be detected by screening or other testing practices, such as those of the prostate, female breast, and colorectum, state variation in incidence rates reflects differences in the use of screening tests or detection practices in addition to differences in disease occurrence.

Cancer Occurrence by Race/Ethnicity

Cancer incidence and death rates vary considerably among racial and ethnic groups (Table 9). For all cancer sites combined, African American men have a 14% higher incidence rate and a 33% higher death rate than white men, whereas African American women have a 6% lower incidence rate but a 17% higher death rate than white women. For the specific cancer sites listed in Table 9, incidence and death rates are consistently higher in African Americans than in whites except for cancers of the breast (incidence) and lung (incidence and mortality) among women, and kidney (mortality) among both men and women. Factors known to contribute to racial disparities in mortality vary by cancer site and include differences in exposure to underlying risk factors (eg, historical smoking prevalence for lung cancer among men), access to high-quality screening

TABLE 7. Cancer Incidence Rates* by Site and State, United States, 2003-2007

STATE	ALL SITES		BREAST	COLORECTUM		LUNG & BRONCHUS		NON-HODGKIN LYMPHOMA		PROSTATE	URINARY BLADDER	
	MALE	FEMALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	MALE	FEMALE
Alabama†	567.5	381.2	114.5	61.4	41.6	106.3	53.2	20.0	13.9	158.4	32.0	7.7
Alaska	512.0	423.5	128.6	56.8	44.3	84.2	63.5	22.1	16.7	133.4	37.5	7.8
Arizona	452.0	355.0	103.6	44.7	33.1	65.4	48.5	18.0	13.1	123.4	33.0	8.4
Arkansas	565.2	386.5	111.3	57.5	41.9	110.9	60.2	22.1	15.2	161.3	33.3	8.6
California	508.9	392.4	121.0	51.4	38.8	63.9	46.3	22.5	15.5	147.1	34.0	8.1
Colorado	503.6	393.8	122.4	49.8	38.6	58.8	45.3	21.7	16.0	158.9	33.3	8.5
Connecticut	589.3	456.3	134.5	59.4	44.4	80.5	60.3	26.0	18.1	163.5	46.3	12.5
Delaware	612.6	443.6	125.7	61.4	44.0	98.0	70.7	23.9	16.6	182.2	43.6	11.8
Dist. of Columbia‡	569.5	421.9	139.4	58.1	47.9	79.4	46.3	22.9	13.4	185.4	24.8	8.6
Florida	532.0	401.0	112.5	53.1	40.4	86.7	59.4	21.5	15.2	137.2	36.4	9.4
Georgia	562.7	393.2	118.5	56.9	41.2	98.8	53.9	21.1	14.3	162.0	32.7	7.9
Hawaii	493.8	386.8	120.6	59.5	40.1	69.2	40.5	19.4	12.4	131.6	25.8	6.5
Idaho	536.2	404.9	116.3	48.2	38.4	68.3	49.1	21.8	17.1	165.8	36.0	9.0
Illinois	576.7	430.3	122.6	65.6	47.3	91.2	59.4	24.2	16.2	157.0	40.2	10.5
Indiana	552.7	416.1	113.8	61.3	45.2	102.4	63.9	22.9	17.0	137.2	37.2	9.4
Iowa	557.2	429.2	122.4	61.9	48.0	89.3	54.2	25.2	18.1	141.8	41.4	9.3
Kansas	559.3	419.3	124.6	60.7	42.4	87.6	53.7	24.3	18.1	158.5	36.2	8.9
Kentucky	610.0	452.8	120.1	67.6	48.9	131.3	78.2	23.5	17.1	141.7	39.2	10.1
Louisiana†	616.4	409.0	118.8	66.7	46.0	107.8	58.9	23.5	16.6	174.5	35.4	8.5
Maine	618.9	466.2	128.8	61.6	47.2	99.1	66.6	24.6	18.8	166.2	49.8	13.9
Maryland‡	537.8	414.7	123.8	54.4	41.4	81.5	57.9	20.9	14.5	159.4	32.8	9.8
Massachusetts	594.0	456.8	131.7	60.5	43.9	82.2	63.1	24.5	16.9	164.6	45.9	12.7
Michigan	591.8	437.2	122.2	57.1	43.4	91.9	62.5	25.7	18.7	173.0	41.9	10.7
Minnesota	567.2	418.4	125.9	54.8	41.6	69.0	49.7	26.3	17.8	183.4	40.0	10.1
Mississippi†,‡	589.5	383.7	109.7	64.1	46.3	114.5	54.9	20.6	13.8	170.8	29.4	7.3
Missouri	549.3	417.8	119.8	61.1	44.0	104.1	63.9	21.8	15.8	132.5	35.7	8.6
Montana	527.8	405.3	120.2	50.3	39.6	74.5	58.3	22.5	14.5	168.5	38.3	9.3
Nebraska	562.4	419.2	122.8	66.6	47.4	84.2	51.2	24.4	17.7	159.0	37.1	9.5
Nevada§	—	—	—	—	—	—	—	—	—	—	—	—
New Hampshire	578.8	454.6	130.1	56.0	43.1	82.5	62.4	23.5	18.1	155.7	46.8	13.3
New Jersey	598.2	451.2	128.4	62.6	46.0	78.3	56.3	25.6	17.7	172.4	46.7	12.1
New Mexico	474.8	365.1	109.3	48.2	35.9	55.7	38.7	18.3	14.3	144.4	26.2	7.3
New York	576.8	435.6	124.3	58.4	44.3	78.2	54.3	25.0	17.5	165.8	42.2	11.1
North Carolina	561.6	406.3	121.4	56.0	40.9	101.0	57.6	21.9	15.4	153.9	35.7	9.0
North Dakota	552.3	410.0	123.4	68.5	43.5	73.6	48.0	23.1	16.8	165.8	40.3	10.4
Ohio	548.4	418.6	119.9	60.0	44.5	96.1	59.7	23.1	16.4	145.5	38.8	9.5
Oklahoma	572.3	428.9	126.8	58.6	43.7	105.3	64.9	23.2	17.8	154.0	35.9	8.8
Oregon	527.1	428.4	130.2	51.8	39.9	77.1	60.1	24.0	16.6	146.8	38.7	9.9
Pennsylvania	590.0	447.4	123.9	63.9	47.4	90.0	57.1	25.0	17.5	158.1	44.9	11.3
Rhode Island	607.1	460.0	130.0	61.8	45.7	92.6	61.9	24.9	17.4	153.5	52.9	13.0
South Carolina	576.5	398.6	119.8	58.5	42.8	100.2	53.7	20.8	14.4	166.5	31.6	8.0
South Dakota	526.0	387.4	116.8	56.6	42.7	77.4	46.3	21.0	16.5	165.0	35.7	7.9
Tennessee‡	543.8	399.1	116.5	57.8	43.0	109.8	60.1	21.5	15.5	135.6	33.4	8.1
Texas†	539.1	389.6	113.3	56.3	39.1	86.0	50.9	22.5	16.0	145.2	30.2	7.3
Utah	483.4	342.4	108.1	44.4	31.6	36.2	23.2	22.6	16.1	178.8	28.8	5.9
Vermont	562.1	456.4	130.4	49.4	42.9	84.5	61.1	23.8	18.3	155.5	45.1	12.6
Virginia	539.1	391.9	122.1	54.2	41.0	88.5	53.8	20.8	13.9	159.1	33.8	8.5
Washington	559.5	436.8	130.3	51.2	38.6	76.1	59.3	27.0	18.2	161.7	40.3	9.8
West Virginia	582.5	439.9	115.3	68.0	48.7	116.3	71.3	24.0	17.3	140.1	39.7	11.0
Wisconsin	543.8	426.3	122.0	54.6	42.2	76.8	53.8	25.5	18.7	148.3	39.7	11.1
Wyoming	512.0	389.9	114.8	51.0	41.6	59.9	48.3	21.3	15.7	167.9	40.8	9.3
United States	552.5	414.7	120.7	57.1	42.4	84.9	55.6	23.2	16.3	153.5	37.7	9.6

*Per 100,000, age adjusted to the 2000 US standard population.

†Due to the effect of large migrations of populations on this state as a result of Hurricane Katrina in September 2005, statistics exclude cases diagnosed in 2005 from July-December.

‡This state is not included in the overall US rates because its registry did not achieve high quality data standards for one or more years during 2003-2007 as determined by the North American Association of Central Cancer Registries (NAACCR).

§This state's registry did not submit incidence data to NAACCR for 2003-2007.

Source: NAACCR, 2010. Data are collected by cancer registries participating in NCI's SEER program and CDC's National Program of Cancer Registries.

TABLE 8. Cancer Death Rates* by Site and State, United States, 2003-2007

STATE	ALL SITES		BREAST	COLORECTUM		LUNG & BRONCHUS		NON-HODGKIN LYMPHOMA		PANCREAS		PROSTATE
	MALE	FEMALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE	FEMALE	MALE
Alabama	263.8	159.9	24.8	23.6	15.1	92.2	41.7	8.6	5.7	12.7	9.2	30.1
Alaska	213.0	156.0	22.7	21.0	13.8	64.8	44.4	7.4	4.8	11.7	9.3	20.6
Arizona	190.6	135.4	21.3	18.5	12.5	54.1	34.9	7.7	4.9	11.0	7.7	21.0
Arkansas	255.5	164.9	24.4	23.4	15.7	93.2	47.5	8.9	5.3	12.6	9.4	27.1
California	198.7	145.2	22.8	18.8	13.5	51.4	34.5	8.3	5.2	11.6	9.3	23.6
Colorado	190.6	139.4	21.3	18.7	13.9	47.8	32.9	8.4	5.0	10.9	8.8	24.5
Connecticut	218.2	154.7	23.7	18.8	14.2	59.6	40.1	8.8	5.5	14.2	10.0	25.5
Delaware	241.4	168.4	24.6	22.3	16.0	76.4	50.4	9.2	5.0	11.2	9.5	26.1
Dist. of Columbia	258.1	162.2	28.3	24.4	17.6	69.1	34.9	8.8	4.1	15.5	10.4	41.7
Florida	211.0	145.3	22.1	18.9	13.4	66.2	40.6	8.3	5.1	11.7	8.5	20.5
Georgia	241.1	153.0	23.9	21.3	14.7	81.6	39.7	8.2	5.0	12.5	9.1	28.9
Hawaii	187.3	122.6	17.9	19.9	11.4	51.2	27.7	7.5	4.3	12.2	9.3	17.1
Idaho	202.3	146.6	21.6	17.2	13.7	53.4	35.4	8.4	6.2	11.5	10.3	27.5
Illinois	235.4	163.4	25.2	23.9	16.5	71.1	42.2	9.2	5.7	13.0	9.9	26.2
Indiana	249.4	166.7	24.5	24.0	15.7	83.9	47.6	10.0	6.0	13.1	9.4	25.6
Iowa	225.4	153.3	22.0	22.1	15.8	70.8	39.1	9.5	5.9	11.7	8.9	25.9
Kansas	225.2	154.1	24.0	21.6	15.0	72.3	41.2	9.7	6.0	12.5	9.4	22.6
Kentucky	275.0	177.6	24.2	25.2	17.6	105.2	56.0	9.6	6.0	12.5	9.4	25.8
Louisiana	270.7	171.5	27.7	26.3	16.9	89.7	45.6	9.4	5.8	13.6	10.7	28.8
Maine	245.0	169.5	22.6	20.8	16.3	76.9	48.7	9.4	5.6	12.9	9.9	25.4
Maryland	230.8	161.7	25.8	22.8	15.6	69.1	42.9	8.1	5.1	12.8	10.5	27.5
Massachusetts	230.5	159.4	22.9	21.1	14.8	65.5	43.8	8.9	5.7	13.4	10.2	24.6
Michigan	232.8	163.1	24.5	21.1	15.4	72.5	44.1	9.6	6.3	13.4	9.7	23.9
Minnesota	211.9	149.1	21.8	18.8	13.7	58.3	37.0	9.5	5.5	11.7	9.1	25.3
Mississippi	276.3	162.0	25.8	24.9	16.9	99.4	43.0	8.3	4.9	13.5	9.8	32.1
Missouri	245.7	164.9	25.8	22.5	15.6	84.3	46.6	8.9	5.6	12.9	9.5	23.6
Montana	211.2	156.5	21.4	18.1	14.3	60.7	43.3	8.7	6.1	12.1	8.8	27.8
Nebraska	218.8	148.0	22.4	23.1	15.7	65.6	35.6	9.1	6.2	12.1	8.4	24.5
Nevada	217.9	165.2	23.9	22.0	16.5	65.0	50.9	7.0	5.3	11.9	9.5	24.5
New Hampshire	227.2	162.2	23.1	21.0	14.8	65.5	44.7	8.7	5.5	12.4	11.2	26.2
New Jersey	222.5	163.2	27.0	23.3	16.7	61.5	39.7	8.9	5.8	13.1	9.8	23.9
New Mexico	194.8	138.0	22.1	19.2	13.3	46.1	29.9	7.4	4.9	11.2	9.1	25.4
New York	206.0	150.5	23.9	20.8	15.0	57.7	36.7	8.0	5.2	12.4	9.6	23.5
North Carolina	244.8	157.7	24.8	21.2	14.5	82.5	42.0	8.3	5.4	12.8	9.6	27.7
North Dakota	210.8	147.9	22.4	21.3	14.8	58.6	35.1	8.4	5.2	11.6	9.5	26.4
Ohio	249.1	168.3	26.6	23.6	16.8	80.3	45.3	9.6	5.9	12.9	9.6	26.3
Oklahoma	246.0	162.6	24.7	23.1	15.0	85.0	47.1	9.3	5.9	11.8	8.5	23.6
Oregon	219.9	161.7	23.2	19.3	14.8	64.2	45.5	9.5	6.2	12.4	10.0	26.0
Pennsylvania	239.6	164.0	25.6	23.6	16.1	71.4	40.4	9.6	6.2	13.4	9.9	25.0
Rhode Island	234.9	158.6	22.8	21.0	14.6	69.8	42.0	8.8	5.2	11.5	9.3	24.2
South Carolina	249.0	156.7	24.4	21.4	15.2	83.9	40.6	8.0	5.2	12.4	9.3	28.9
South Dakota	220.3	145.7	22.3	21.4	15.1	65.5	36.5	8.8	5.4	11.2	9.5	26.0
Tennessee	264.0	167.1	25.4	23.1	15.9	95.7	47.5	9.5	5.9	12.7	9.2	27.6
Texas	221.3	147.3	23.0	21.0	13.9	68.3	37.5	8.3	5.3	11.6	8.6	23.1
Utah	161.7	116.3	22.8	15.1	10.9	31.4	17.6	8.1	5.2	9.8	8.1	25.7
Vermont	215.3	156.7	23.5	20.5	15.4	61.8	42.5	8.5	5.1	10.8	8.9	25.2
Virginia	235.7	158.0	25.6	21.7	14.7	74.5	42.1	8.2	5.3	13.0	9.8	27.3
Washington	214.0	158.5	23.0	18.3	13.4	61.5	44.3	9.0	5.8	12.2	9.6	25.4
West Virginia	259.4	175.8	24.3	25.3	17.9	91.0	50.5	10.0	6.4	11.4	7.5	22.3
Wisconsin	223.5	154.5	22.6	19.9	14.0	62.3	38.5	9.3	6.0	12.6	9.4	27.1
Wyoming	204.6	155.2	23.4	20.1	16.0	55.6	38.4	8.3	6.8	12.3	10.7	21.8
United States	225.4	155.4	24.0	21.2	14.9	68.8	40.6	8.7	5.5	12.3	9.4	24.7

*Per 100,000, age adjusted to the 2000 US standard population.

Source: US Mortality Data, National Center for Health Statistics, Centers for Disease Control and Prevention.

TABLE 9. Incidence and Death Rates* by Site, Race, and Ethnicity, United States, 2003-2007

	WHITE	AFRICAN AMERICAN	ASIAN AMERICAN AND PACIFIC ISLANDER	AMERICAN INDIAN AND ALASKA NATIVE†	HISPANIC/LATINO‡
Incidence					
All sites					
Male	544.9	623.1	332.3	424.6	426.1
Female	418.8	392.9	278.1	359.2	331.2
Breast (female)	121.9	114.6	82.3	88.2	91.0
Colorectum					
Male	56.1	67.2	42.8	51.9	49.2
Female	41.4	50.7	32.5	42.2	34.9
Kidney & renal pelvis					
Male	20.2	21.6	9.6	26.9	18.9
Female	10.6	11.0	4.7	16.5	11.0
Liver & bile duct					
Male	8.2	13.5	21.6	14.3	16.4
Female	2.8	3.9	8.1	7.2	6.2
Lung & bronchus					
Male	84.3	103.5	49.9	70.2	48.0
Female	57.0	51.8	27.7	50.6	27.1
Prostate	143.8	230.0	81.0	101.5	128.0
Stomach					
Male	8.7	16.4	17.2	14.5	14.1
Female	4.1	8.4	9.7	7.3	8.6
Uterine cervix	7.7	10.7	7.4	9.7	12.5
Mortality					
All sites					
Male	222.5	296.5	134.2	183.7	150.6
Female	155.0	180.6	94.1	138.0	102.3
Breast (female)	23.4	32.4	12.2	17.6	15.3
Colorectum					
Male	20.6	30.5	13.2	19.2	15.6
Female	14.4	21.0	9.9	12.9	10.5
Kidney & renal pelvis					
Male	6.0	6.0	2.6	8.8	5.2
Female	2.7	2.7	1.2	3.8	2.4
Liver & bile duct					
Male	7.0	11.1	14.7	10.9	11.3
Female	3.0	3.9	6.4	6.6	5.2
Lung & bronchus					
Male	68.3	87.5	36.7	48.1	32.5
Female	41.6	39.6	18.5	33.3	14.4
Prostate	22.8	54.2	10.6	20.0	18.8
Stomach					
Male	4.6	10.7	9.4	9.2	8.0
Female	2.4	5.0	5.6	4.2	4.6
Uterine cervix	2.2	4.4	2.1	3.4	3.1

*Per 100,000 population, age adjusted to the 2000 US standard population.

†Data based on Contract Health Service Delivery Areas, comprising about 55% of the US American Indian/Alaska Native population; for more information please see: Espey et al.²³

‡Persons of Hispanic/Latino origin may be of any race.

Source: Kohler, et al.¹²

(breast, cervical, and colorectal cancers), and timely diagnosis and treatment for many cancers. The higher breast cancer incidence rates noted among white women are thought to reflect a combination of factors that affect both diagnosis (historically more prevalent mammography use in white women) and underlying disease occurrence (such as later age at first birth and

greater use of menopausal hormone therapy among white women).²²

Cancer incidence and death rates are lower in other racial and ethnic groups than in whites and African Americans for all cancer sites combined and for the 4 most common cancer sites. However, incidence and death rates for cancer sites related to

TABLE 10. Fixed-Interval Trends in Overall Cancer Incidence and Death Rates by Race/Ethnicity, United States, 1998 to 2007

	1998 - 2007 AAPC			
	INCIDENCE		MORTALITY	
	MALE	FEMALE	MALE	FEMALE
All race/ethnicities combined	-0.8*	-0.5*	-1.8*	-1.1*
White	-0.9*	-0.5*	-1.7*	-1.0*
African American	-1.4*	-0.5	-2.6*	-1.4*
Asian American/Pacific Islander	-1.4*	-0.3*	-2.0*	-1.2*
American Indian/Alaska Native†	-1.3*	-0.2	-1.0*	-0.2
Hispanic‡	-1.4*	-0.6*	-2.5*	-1.3*

AAPC indicates average annual percent change. Race and ethnicity categories are not mutually exclusive of Hispanic origin.

*AAPC is statistically significantly different from zero (two-sided Z test, $P < 0.05$).

†Data based on Contract Health Service Delivery Areas.

‡Excludes deaths from the District of Columbia, Maine, Minnesota, New Hampshire, and North Dakota.

Source: Kohler, et al.¹²

infectious agents, such as those of the uterine cervix, stomach, and liver, are generally higher in minority populations than in whites. Stomach and liver cancer incidence and death rates are at least twice as high in Asian Americans/Pacific Islanders compared with whites, reflecting an increased prevalence of chronic infection with *Helicobacter pylori* and hepatitis B and C viruses in this population. Kidney cancer incidence and death rates are the highest among American Indians/Alaska Natives; the higher prevalence of obesity and smoking in this population are factors that may contribute to this disparity.²³

Trends in cancer incidence can be adjusted for delayed reporting only in whites and African Americans because the long-term incidence data required for delay adjustment are not available for other racial and ethnic subgroups. From 1998 to 2007, both incidence (unadjusted for delayed reporting) and death rates for all cancer sites combined decreased among men of all racial/ethnic groups; the largest declines in death rates occurred among men of African American (2.6% per year) and Hispanic heritage (2.5% per year) (Table 10).¹² It is important to note, however, that cancer death rates in African American men remain substantially higher than those among white men and twice those of Hispanic men. Among women during this time period, incidence rates decreased in whites, Asian Americans/Pacific Islanders, and Hispanics, and were

stable in African Americans and American Indians/Alaska Natives; cancer death rates decreased among women of all racial/ethnic groups except American Indians/Alaska Natives, in whom they were stable.

The Impact of Eliminating Disparities on Premature Death From Cancer

Level of education is often used as a marker of socioeconomic status. In 2007, cancer death rates in the least educated segment of the population were 2.6 times higher than those in the most educated segment (Table 11). The racial patterns in educational disparities were generally similar across the 4 major cancer sites, although the magnitude of the association was generally weaker for Hispanics. The largest socioeconomic disparity was seen for lung cancer; the death rate in men was 5 times higher for the least educated than for the most educated. Differences in lung cancer death rates reflect the striking gradient in smoking prevalence by level of education; 31% of men with 12 or fewer years of education are current smokers, compared with 12% of college graduates and 5% of men with graduate degrees.²⁴

Figure 8 shows the numbers of potential premature cancer deaths that could have been avoided in 2007 among adults aged 25 to 64 years in the absence of socioeconomic and/or racial disparities. If everyone in the United States experienced the same overall cancer death rates as the most educated non-Hispanic whites, 37% (60,370 of 164,190) of the premature cancer deaths could potentially have been avoided. This analysis suggests that eliminating socioeconomic disparities in African Americans could potentially avert twice as many premature cancer deaths as eliminating racial disparities, underscoring the dominant role of poverty in cancer disparities.

Cancer Survival By Race

Compared with whites, African American men and women have poorer survival once cancer is diagnosed. The 5-year relative survival is lower in African Americans than in whites for every stage of diagnosis for nearly every cancer site (Fig. 9). These disparities may result from inequalities in access to and receipt of quality health care and/or from differences in comorbidities. As shown in Figure 10, African Americans are less likely than whites to be diagnosed with cancer at a localized stage, when the disease may be more

TABLE 11. Cancer Death Rates* by Educational Attainment, Race, and Sex, United States, 2007

	MEN				WOMEN			
	ALL RACES	NON-HISPANIC AFRICAN AMERICAN	NON-HISPANIC WHITE	HISPANIC	ALL RACES	NON-HISPANIC AFRICAN AMERICAN	NON-HISPANIC WHITE	HISPANIC
All sites								
< = 12 years of education	147.85	216.48	148.79	52.80	119.38	145.38	123.96	55.99
13-15 years of education	72.67	101.67	71.33	45.71	69.07	105.88	66.24	35.84
>= 16 years of education	55.92	76.90	56.48	37.05	59.13	86.18	57.79	58.68
RR (95% CI)	2.64 (2.53 - 2.76)	2.82 (2.40 - 3.30)	2.63 (2.52 - 2.76)	1.43 (1.06 - 1.92)	2.02 (1.94 - 2.10)	1.69 (1.49 - 1.90)	2.15 (2.05 - 2.25)	0.95 (0.69 - 1.32)
Absolute difference	91.94	139.58	92.32	15.75	60.25	59.20	66.17	-2.68
Lung								
< = 12 years of education	51.63	73.01	53.49	9.40	33.86	33.20	37.71	5.43
13-15 years of education	20.54	28.26	20.48	6.85	15.28	20.22	15.29	4.42
>= 16 years of education	10.35	17.64	10.18	8.61	8.77	11.96	8.62	6.48
RR (95% CI)	4.99 (4.65 - 5.34)	4.14 (3.27 - 5.24)	5.26 (4.88 - 5.67)	1.09 (0.66 - 1.82)	3.86 (3.58 - 4.17)	2.78 (2.22 - 3.48)	4.38 (4.02 - 4.76)	0.84 (0.38 - 1.83)
Absolute difference	41.28	55.37	43.31	0.79	25.09	21.24	29.09	-1.05
Colorectum								
< = 12 years of education	13.59	22.45	13.18	5.34	9.75	13.97	9.74	5.11
13-15 years of education	7.41	13.46	6.74	6.30	5.65	9.87	5.23	3.26
>= 16 years of education	6.22	10.37	6.05	3.80	4.73	9.81	4.43	4.60
RR (95% CI)	2.18 (2.00 - 2.39)	2.17 (1.66 - 2.87)	2.18 (1.97 - 2.41)	1.41 (0.67 - 2.96)	2.06 (1.86 - 2.29)	1.42 (1.11 - 1.83)	2.20 (1.95 - 2.48)	1.11 (0.47 - 2.60)
Absolute difference	7.37	12.08	7.13	1.54	5.02	4.16	5.31	0.51
Prostate								
< = 12 years of education	3.61	9.03	3.04	1.33				
13-15 years of education	2.16	5.51	1.81	1.85				
>= 16 years of education	2.17	5.99	2.05	0.82			NA	
RR (95% CI)	1.66 (1.44 - 1.93)	1.51 (1.03 - 2.22)	1.48 (1.25 - 1.75)	1.61 (0.36 - 7.20)				
Absolute difference	1.44	3.04	0.99	0.51				
Breast								
< = 12 years of education					22.12	33.53	21.41	11.93
13-15 years of education					16.23	31.17	14.60	7.97
>= 16 years of education			NA		16.51	27.44	15.76	18.46
RR (95% CI)					1.34 (1.26 - 1.43)	1.22 (1.03 - 1.44)	1.36 (1.26 - 1.46)	0.65 (0.41 - 1.03)
Absolute difference					5.60	6.09	5.64	-6.52

RR indicates relative risk of cancer death among those with the lowest level of education compared to those with the highest level; CI, confidence interval; NA, not applicable. Absolute difference is between the lowest and highest education levels.

Education categories are defined based on 1989 death certificates.

* Rates are for individuals 25-64 years at death, per 100,000, and age-adjusted to the 2000 US standard population.

Source: National Center for Health Statistics

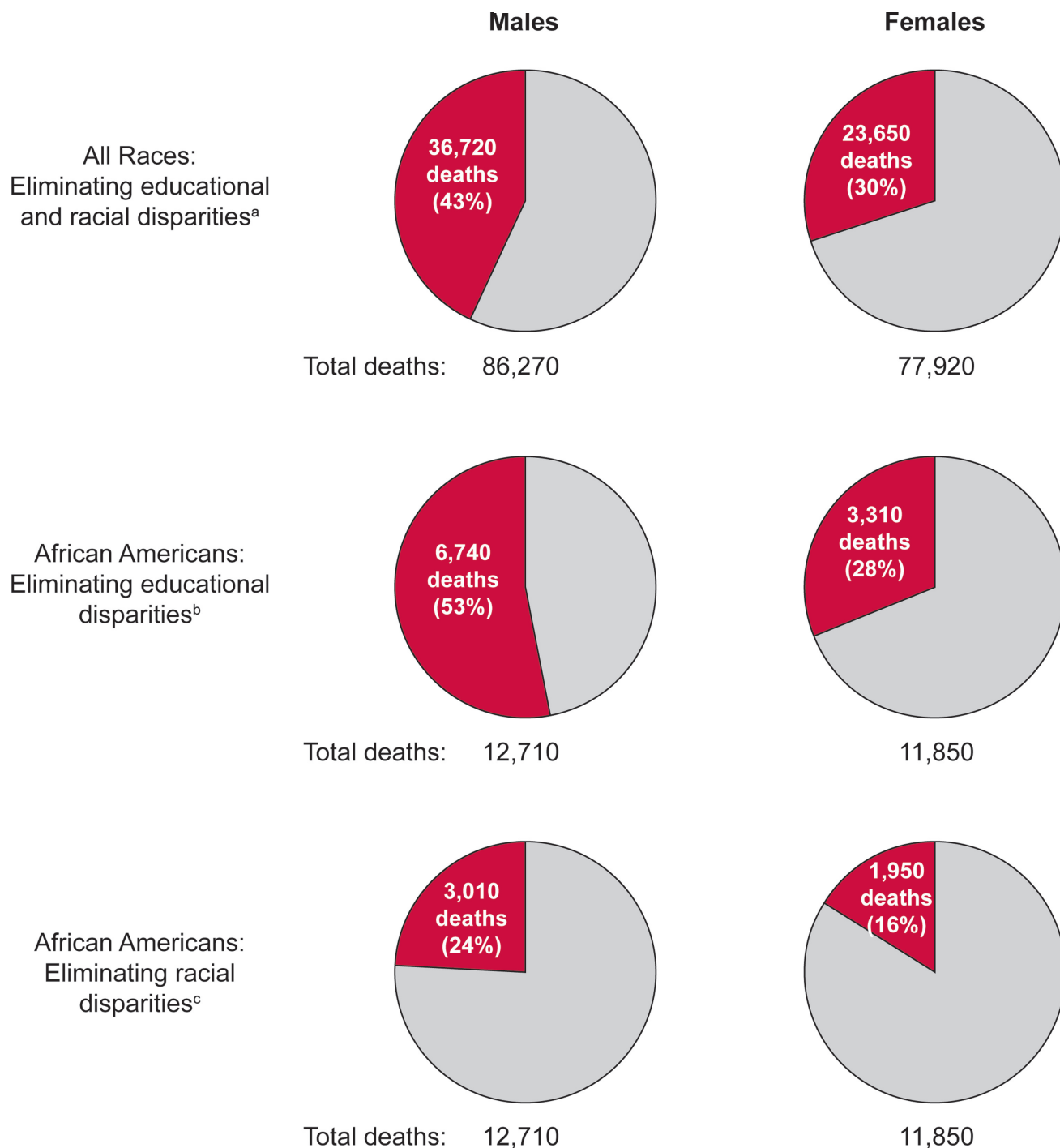


FIGURE 8. Potential US* Cancer Deaths That Could Have Been Avoided by Eliminating Educational and/or Racial Disparities, Aged 25 to 64 Years, 2007.

*Excludes Rhode Island and Georgia.

^aAge-specific cancer death rates of the most educated non-Hispanic whites in 2007 were applied to all races.

^bAge-specific cancer death rates of the most educated African Americans in 2007 were applied to all African Americans.

^cAge- and educational attainment-specific cancer death rates of non-Hispanic whites in 2007 were applied to the corresponding population of African Americans.

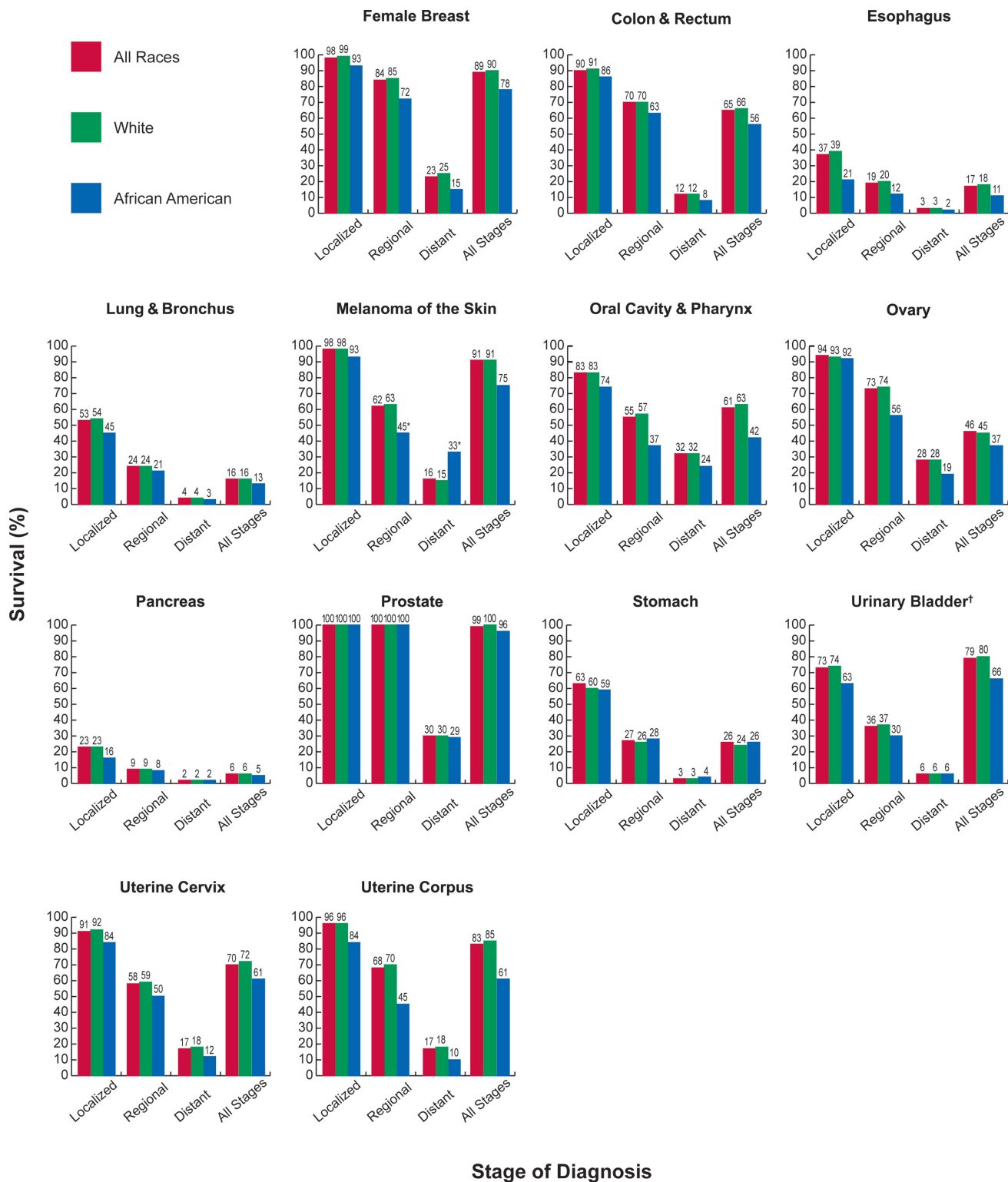


FIGURE 9. Five-Year Relative Survival Rates Among Patients Diagnosed With Selected Cancers by Race and Stage at Diagnosis, United States, 1999 to 2006.

*The standard error of the survival rate is between 5 and 10 percentage points.

†The survival rate for in situ urinary bladder cancer is 97% for All Races and Whites and 94% for African Americans. Staging was performed according to Surveillance, Epidemiology, and End Results (SEER) historic stage categories rather than the American Joint Committee on Cancer (AJCC) staging system. Source: Altekruse, et al.²

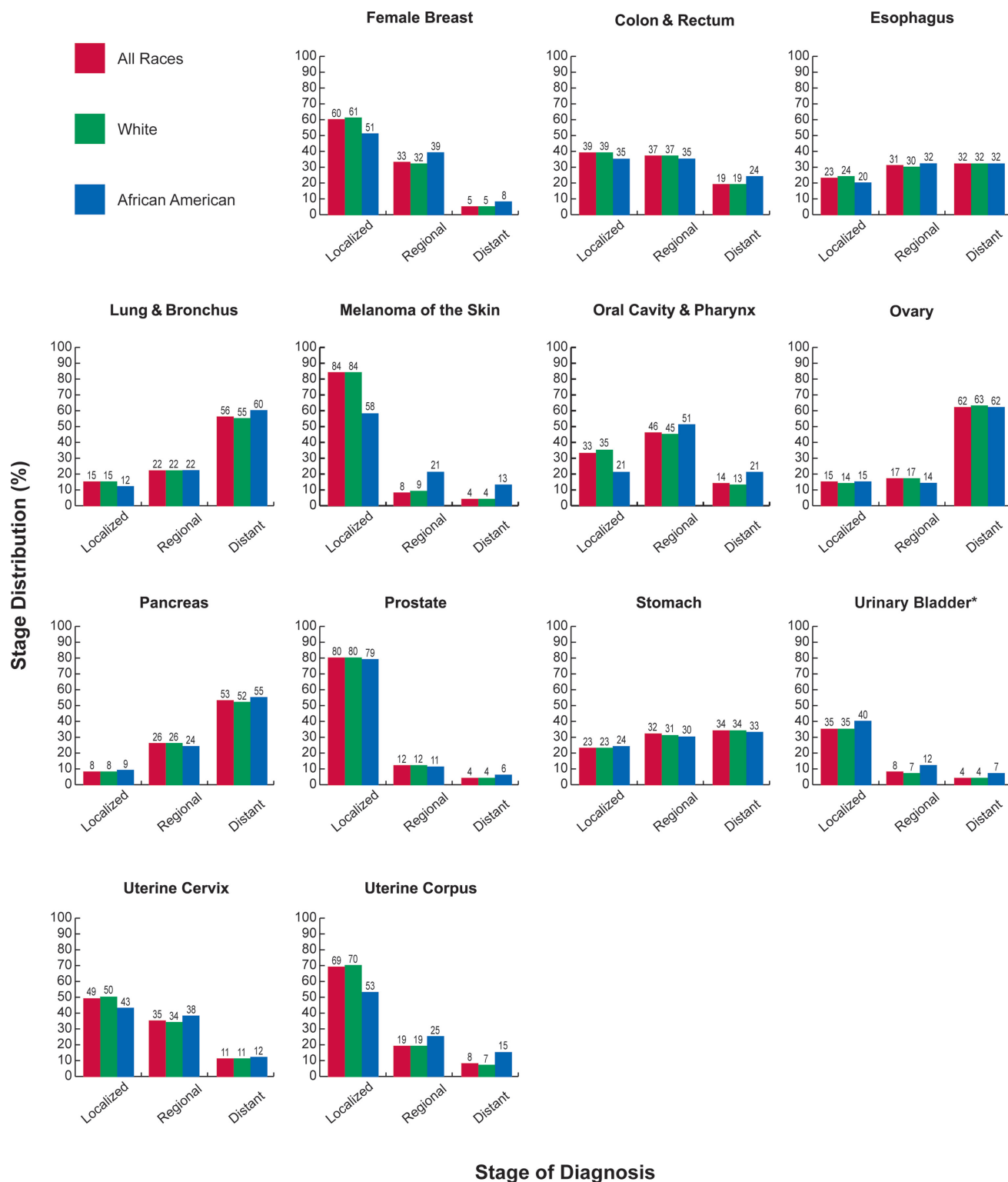


FIGURE 10. Distribution of Selected Cancers by Race and Stage at Diagnosis, United States, 1999 to 2006.

*The proportion of in situ urinary bladder cancer cases is 50%, 51%, and 37% in All Races, Whites, and African Americans, respectively. Staging was performed according to Surveillance, Epidemiology, and End Results (SEER) historic stage categories rather than the American Joint Committee on Cancer (AJCC) staging system. For each cancer type, stage categories do not total 100% because sufficient information is not available to assign a stage to all cancer cases. Source: Altekruse, et al.²

TABLE 12. Trends in 5-Year Relative Survival Rates* (%) by Race and Year of Diagnosis, United States, 1975 to 2006

	ALL RACES			WHITE			AFRICAN AMERICAN		
	1975 TO 1977	1984 TO 1986	1999 TO 2006	1975 TO 1977	1984 TO 1986	1999 TO 2006	1975 TO 1977	1984 TO 1986	1999 TO 2006
All sites	50	54	68†	51	55	69†	40	41	59†
Brain	24	29	36†	23	28	35†	27	32	41†
Breast (female)	75	79	90†	76	81	91†	62	65	78†
Colon	52	59	66†	52	60	67†	47	50	55†
Esophagus	5	10	19†	6	11	20†	3	9	13†
Hodgkin lymphoma	74	80	87†	74	80	88†	71	75	82†
Kidney	51	56	70†	51	56	70†	50	54	67†
Larynx	67	66	63†	68	68	65	59	53	49†
Leukemia	36	42	55†	36	43	56†	34	34	47†
Liver & bile duct	4	6	14†	4	6	14†	2	5	10†
Lung & bronchus	13	13	16†	13	14	17†	12	11	13†
Melanoma of the skin	83	87	93†	83	87	93†	60‡	70§	74‡
Myeloma	26	29	39†	26	27	39†	31	32	38†
Non-Hodgkin lymphoma	48	53	69†	49	54	71†	49	48	60†
Oral cavity	53	55	63†	55	57	65†	36	36	45†
Ovary	37	40	45†	37	39	45†	43	41	37
Pancreas	3	3	6†	3	3	6†	2	5	5†
Prostate	69	76	100†	70	78	100†	61	66	97†
Rectum	49	57	69†	50	58	70†	45	46	60†
Stomach	16	18	27†	15	18	26†	16	20	26†
Testis	83	93	96†	83	93	97†	73‡,#	87‡	87
Thyroid	93	94	97†	93	94	98†	91	90	95
Urinary bladder	74	78	81†	75	79	82†	51	61	66†
Uterine cervix	70	68	71	71	70	73	65	59	64
Uterine corpus	88	84	84†	89	85	86†	61	58	61

*Survival rates are adjusted for normal life expectancy and are based on cases diagnosed in the SEER 9 areas from 1975-77, 1984-86, and 1999-2006 and followed through 2007.

†The difference in rates between 1975-1977 and 1999-2006 is statistically significant ($p < 0.05$).

‡The standard error of the survival rate is between 5 and 10 percentage points.

§The standard error of the survival rate is greater than 10 percentage points.

#Survival rate is for 1978-1980.

Source: Altekruse, et al.²

TABLE 13. Trends in Five-Year Relative Survival Rates* (%) for Children Under Age 15, United States, 1975 to 2006

SITE	YEAR OF DIAGNOSIS								
	1975 TO 1977	1978 TO 1980	1981 TO 1983	1984 TO 1986	1987 TO 1989	1990 TO 1992	1993 TO 1995	1996 TO 1998	1999 TO 2006
All sites	58	63	67	68	72	76	77	79	82†
Acute lymphocytic leukemia	58	66	71	73	78	83	84	87	89†
Acute myeloid leukemia	19	26	27‡	31‡	37‡	42	42‡	49	61†
Bone & joint	50‡	48	57‡	58‡	67‡	67	74	70	73†
Brain & other nervous system	57	58	56	62	64	65	70	75	74†
Hodgkin lymphoma	81	88	88	91	87	97	95	96	96†
Neuroblastoma	53	57	55	52	63	76	67	66	72†
Non-Hodgkin lymphoma	44	53	67	70	71	77	81	83	87†
Soft tissue	61	75	69	73	66	80	77	70	81†
Wilms tumor	73	79	87	91	92	92	92	92	92†

*Survival rates are adjusted for normal life expectancy and are based on follow-up of patients through 2007.

†The difference in rates between 1975 to 1977 and 1999 to 2006 is statistically significant ($p < 0.05$).

‡The standard error of the survival rate is between 5 and 10 percentage points.

Source: Altekruse, et al.²

easily and successfully treated. The extent to which factors other than stage at diagnosis contribute to the overall survival differential is unclear.²⁵ However, some studies suggest that African Americans who receive cancer treatment and medical care similar to that of whites experience similar outcomes.²⁶

There have been notable improvements since 1975 in the relative 5-year survival rates for many cancer sites and for all cancers combined for both whites and African Americans (Table 12). The improvement in survival reflects a combination of earlier diagnosis and improved treatments. Cancers for which survival has shown the least improvement over the past 30 years include lung and pancreas.

Relative survival rates cannot be calculated for all racial and ethnic subpopulations because accurate life expectancies are not available. However, based on cause-specific survival rates of cancer patients diagnosed from 1999 to 2006 in SEER areas of the United States, all minority male populations have a greater probability of dying from cancer within 5 years of diagnosis than whites. Among women, African Americans have the lowest 5-year cancer-specific survival (56.0%), followed by American Indians/Alaska Natives (60.6%), Hispanics (65.5%), whites (65.6%), and Asian Americans/Pacific Islanders (68.2%).² For the 4 major cancer sites (prostate, female breast, lung and bronchus, and colorectum), minority populations are generally more likely than non-Hispanic whites to be diagnosed at a distant stage of disease.²⁷

Cancer in Children

Cancer is the second most common cause of death among children between the ages of 1 and 14 years in the United States, surpassed only by accidents. About one-third of the invasive cancers diagnosed in children aged birth to 14 years are leukemias (particularly acute lymphocytic leukemia), followed by cancer of the brain and other nervous system (27%), soft tissue sarcomas (7%, half of which are rhabdomyosarcoma), neuroblastoma (7%), renal (Wilms) tumors (5%), and Hodgkin and non-Hodgkin lymphomas (4% each).² Between 1998 and 2007, the overall incidence rate for cancer in children aged 14 years and younger increased by 0.6% per year while

the death rate decreased by 1.0% per year.¹² Over the past 25 years, there have been significant improvements in the 5-year relative survival rate for all of the major childhood cancers due to new and improved treatments (Table 13). The 5-year relative survival rate among children for all cancer sites combined improved from 58% for patients diagnosed between 1975 and 1977 to 82% for those diagnosed between 1999 and 2006.²

Limitations

Estimates of the expected numbers of new cancer cases and cancer deaths should be interpreted cautiously because these estimates are model-based and may vary considerably from year to year for reasons other than changes in cancer occurrence. For instance, estimates are invariably affected by changes in method, which occur regularly as modeling techniques improve over time. For example, the introduction of a new method for projecting incident cancer cases in 2007 substantially affected the estimates for a number of cancers, particularly leukemia and female breast.⁸ In addition, not all changes in cancer trends can be captured by modeling techniques. For these reasons, we discourage the use of these estimates to track year-to-year changes in cancer occurrence and death. The preferred data sources for tracking cancer trends are age-standardized or age-specific cancer death rates from the NCHS and cancer incidence rates from SEER or NPCR, even though these data are 3 and 4 years old, respectively, by the time that they become available. Nevertheless, the American Cancer Society projections of the number of new cancer cases and deaths provide a reasonably accurate estimate of the current cancer burden in the United States.

Errors in reporting race/ethnicity in medical records and on death certificates may result in underestimates of cancer incidence and mortality rates in nonwhite and non-African American populations. It is also important to note that cancer data in the United States are primarily reported for broad racial and ethnic minority groups that are not homogeneous, and thus important difference in the cancer burden within racial/ethnic subgroups are often masked. ■

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